What more cases buy: inferential targets for reproduction-number estimates and what cross-sectional dispersion diagnostics measure

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4 November 2026

Abstract

Case counts often set the precision attributed to reproduction-number estimates, but the error that more cases remove depends on the inferential target. In a one-generation branching model with negative binomial offspring and binomial reporting, we derive exact mean squared errors for the current reproduction number, the realised mean over observed periods and the process mean. Counting error falls inversely with expected count for all three targets; the process mean also retains between-period transmission variance. Using the previous observed count as denominator raised the counting error 1.5-fold at large counts and 2.4-fold at ten expected cases under illustrative parameters. Under period-level reporting fluctuation, single-period count moments cannot separate reporting volatility from transmission variation. A simulation study on the design of 1,748 US state-weeks of influenza hospital admissions (2022–2025) assessed cross-sectional dispersion diagnostics against known truth. The intercept of an affine fit of relative squared deviation on inverse count recovered transmission variance under fixed reporting and the transmission-plus-reporting total under reporting shocks, and was smaller than transmission variance when heterogeneity scaled with count. The observed intercept was 0.0150 (95% CI 0.0109–0.0191 with a two-way cluster-robust sandwich and 0.0101–0.0206 with a conservative bootstrap). Across the full (ρ, k) parameter manifold compatible with the empirical counting slope, implied transmission relative standard deviation spans 11.6–12.3% (9.9–13.8% propagated) under pure heterogeneity, expanding to a global envelope of 5.0–20.4% across mechanisms with reporting shocks and scale dependence. None of these mechanisms reproduced every tested summary under 90%, 95% or 99% decision rules. The counting slope was 1.26 (bootstrap 0.81–1.73) times the independent-Poisson reference and the dispersion-to-counting crossover 37 weekly admissions (sandwich 24–58; bootstrap 18–72). Non-parametric calibrations of the two-way cluster-robust season test on the empirical data (Wild cluster bootstrap $p = 0.013$; permutation $p = 0.045$) reject equal season intercepts at the 5% level, corroborating seasonal transmission divergence. Counting slopes and thinning did not distinguish overdispersion, under-reporting and heterogeneous transmission, whereas in-sample and held-out curve shapes weighed against heterogeneity that does not depend on count.

Keywords: reproduction number; surveillance; overdispersion; reporting variation; simulation study

1 Introduction

Reproduction numbers summarise transmission under specified conditions and are routinely estimated from surveillance counts [1, 2]. Renewal methods relate incidence to past infectiousness through a generation-interval distribution, and the width of their credible intervals shrinks as the number of cases grows [3, 4]. Work on low incidence, reporting noise and timely detection has made the limits of period-specific inference explicit [5–7], and temporal aggregation and reporting delays separate the reproduction number of the transmission process from the apparent reproduction number of an observed series [8–10].

Precision guidance for renewal estimators has been stated in cases. Cori et al. [3] tabulate the number of incident cases a time window needs for a chosen posterior coefficient of variation and recommend starting estimation once at least 12 cases have accrued. Under a gamma prior with shape a_0, that coefficient is $(a_0 + I_W)^{-1/2}$, where I_W is the number of cases in the window, so it is a function of the window count alone. It measures counting error for the average transmission level over the window. The estimate is also often read as the transmission level now or as a typical level for the population. For those targets, variation between periods is an additional error component that more cases do not remove. The criterion is therefore correct for one target and incomplete for the other two, and the distinction depends on which target the analyst intends.

The variance structure behind this distinction is classical. Variance-component models split the variance of a unit-level estimate into a within-unit sampling term and a between-unit term [11, 12]. Random-effects meta-analysis estimates the between-study component on top of within-study error [13], and small-area estimation and empirical-Bayes disease mapping weight each direct estimate against a common mean according to the ratio of the two terms [14–16]. Funnel plots for institutional comparison draw control limits from the counting term and add an overdispersion component when units differ beyond chance [17]. Population ecology makes the same split between demographic and environmental stochasticity

[18, 19], as do stochastic epidemic models [20, 21]. In each setting the count at which the two terms are equal is where a shrinkage estimator places half its weight on the direct estimate.

Several results used below are standard and are cited where they are applied. The mean squared error decompositions follow from the law of total variance [11, 22]. The efficiency of the count-to-scale ratio for a Poisson mean is the Cramér–Rao bound [23], the delta-method expansion of a ratio estimator is textbook material [22], and the square-root allocation of surveillance effort is Neyman allocation [12]. Offspring overdispersion follows Lloyd-Smith et al. [24], and its estimation from small samples Lloyd-Smith [25]. A variance that scales as a power of the mean is Taylor's law [26], the alternative to the affine form that Section 4 tests. The distinction between the realised quantity of a finite population and the parameter of the process that generated it follows Graubard and Korn [27]. While these mathematical decompositions apply classical principles of variance analysis and survey sampling, the operational estimand linking empirical surveillance dispersion to transmission versus reporting mechanisms, the formalization of the crossover count $m_\times$ as an actionable shrinkage balance point, and the known-truth simulation audit on the real design are the distinct contributions of this work.

The paper makes four contributions. Section 2 sets out three inferential targets for a period-specific reproduction number and derives their exact mean squared errors in a branching model with negative binomial offspring and binomial reporting, together with the correction required when the denominator is the previous observed count. Section 3 shows that, when the reporting probability fluctuates between periods, single-period second moments of the counts cannot separate reporting volatility from transmission variation, and that the realised-mean target then retains a reporting term. Section 4 defines three cross-sectional dispersion diagnostics for surveillance counts, with reference values for the counting slope under independent Poisson and branching Poisson counting. The same section reports a known-truth simulation study on the real sampling design, including a grid over the share of variance carried by transmission, the coverage of dependence-aware intervals and simulated reference distributions for the season and held-out comparisons. Section 5 applies the diagnostics to 1,748 state-weeks of US influenza hospital admissions.

The diagnostics address a practical gap. Routine hospital surveillance does not observe the variables of a branching experiment. Admissions follow infection after a delay, aggregate heterogeneous populations and are recorded on a calendar schedule, and their count scale is an outcome of the epidemic. Reporting probabilities may change for all infections in the same period [28, 29], and separating process from observation variation in time series is a known difficulty [30, 31]. The cross-sectional spread of growth-rate or reproduction-number estimates has been read as variation in transmission or behaviour between places and periods without subtracting the part of that spread that depends on case counts [32, 33], although per-unit standard errors make the subtraction possible [34]. Regressing the squared relative deviation of an apparent growth statistic on inverse case count across units, and reading the intercept as transmission variance or an error floor and the slope as counting noise, applies the classical decomposition to such data. Whether those readings hold has to be checked against data with a known truth, and Section 4 reports that check. Section 6 discusses real-time use, reporting that depends on transmission and the choice of generation interval. Proofs, extensions and supplementary analyses, including a cross-regime check on COVID-19 counts from mainland China, are in the appendices.

2 Model and inferential targets

2.1 A branching experiment with a known observation scale

Consider one generation of transmission with $n \geq 1$ primary infections. Conditional on the reproduction number $R_t > 0$, the offspring counts of primary cases are independent negative binomial variables [24]:

$$X_{i,t} \mid R_t \sim \mathrm{NB}(k,\, k/(k+R_t)), \qquad \mathrm{E}(X_{i,t} \mid R_t) = R_t, \quad \mathrm{Var}(X_{i,t} \mid R_t) = R_t + \frac{R_t^2}{k}, \tag{1}$$

where $k > 0$ controls individual heterogeneity. Each secondary infection is independently reported with probability $\rho \in (0, 1]$. Summation over primary cases and binomial thinning give the law of the reported count:

$$C_t \mid n, R_t \sim \mathrm{NB}(nk,\, k/(k+\rho R_t)), \qquad \mathrm{E}(C_t \mid n, R_t) = \rho n R_t, \qquad \mathrm{Var}(C_t \mid n, R_t) = \rho n R_t + \frac{\rho^2 n R_t^2}{k}. \tag{2}$$

Define $m = \rho n$ and $\hat{R}_t = C_t/m$. The scale m is the expected reported count at $R_t = 1$, not the realised count C_t.

The baseline assumptions are fixed, known ρ with independent reporting (H1); conditionally independent offspring as in Eq. (1) (H2); fixed, known k (H3); and a fixed or exogenous n that does not depend on transmission (H4). Write $\mathrm{E}(R_t) = \overline{R} > 0$ and $\mathrm{Var}(R_t) = v_R < \infty$. Across periods, counts are independent given the whole transmission trajectory, each with conditional law (2). These are repeated branching experiments; they do not assert that naturally evolving incidence is exogenous. Dividing by an assumed $\rho_0 \neq \rho$ gives conditional mean $(\rho/\rho_0)R_t$, a multiplicative bias that more cases do not remove.

The conditional counting variance combines offspring randomness and omission. Conditioning on the total number of secondary infections splits it into $(R_t + R_t^2/k)/n$ and $(1-\rho)R_t/(\rho n)$. The term linear in R_t is therefore not purely reporting noise; it remains under complete reporting.

Table 1 lists the symbols used throughout.

Table 1 Notation

Symbol	Meaning
R_t	Reproduction number in period t (current target)
$\overline{R}$, v_R	Mean and variance of R_t under the stated transmission law (process mean and between-period transmission variance)
$\overline{R}_T$	Mean of $R_1, \ldots, R_T$ over the observed periods (realised mean)
n, ρ, k	Primary infections, reporting probability, negative binomial dispersion
$m = \rho n$	Expected reported count at $R_t = 1$ (observation scale)
C_t, $\hat{R}_t = C_t/m$	Reported secondary count and the count-to-scale estimator
M_t, $\tilde{R}_t = C_t/M_t$	Reported primary count and the observed-denominator estimator
A	Counting-error numerator, $\overline{R} + \rho(\overline{R}^2 + v_R)/k$
$m_\times = A/v_R$	Crossover scale at which counting error equals v_R
$\gamma(m)$	Weight of the direct estimate in the best linear predictor
p_t, τ^2	Period-level reporting probability and its squared relative variation
v_H, A_H	Reporting-volatility variance and counting numerator under H5
a, b	Intercept and inverse-count slope of the affine dispersion diagnostic
b_P	Independent-Poisson reference for the slope

2.2 Current values, realised means and process means

For the current target, $\mathrm{E}(\hat{R}_t \mid R_t) = R_t$ and $\mathrm{Var}(\hat{R}_t \mid R_t) = (R_t + \rho R_t^2/k)/m$, which attains the Cramér–Rao bound for estimators unbiased for every R_t under this likelihood (Appendix A.1).

Proposition 1 (One period, two targets)**.** *Under H1–H4, let $A = \overline{R} + \rho(\overline{R}^2 + v_R)/k$. The two risks of $\hat{R}_t$ are:*

$$\mathrm{E}[(\hat{R}_t - R_t)^2] = \frac{A}{m}, \qquad V(m) := \mathrm{E}[(\hat{R}_t - \overline{R})^2] = v_R + \frac{A}{m}. \tag{3}$$

Among estimators based on this period's count that are conditionally unbiased for R_t, $\hat{R}_t$ minimises both risks.

The proof (Appendix A.2) squares the error decomposition:

$$\hat{R}_t - \overline{R} = (\hat{R}_t - R_t) + (R_t - \overline{R}), \tag{4}$$

The two terms on the right are uncorrelated. Between-period transmission variation is part of the signal for the current value and sampling error for the process mean, which is a mean under a stated stationary law and not automatically a seasonal average of a nonstationary epidemic.

For $v_R > 0$ define the crossover scale $m_\times = A/v_R$, the expected count at which the two terms of $V(m)$ are equal. Its meaning is the one it has in small-area estimation [14]. The best linear predictor of R_t from $\hat{R}_t$ under the stated moments takes the form:

$$\overline{R} + \gamma(m)(\hat{R}_t - \overline{R}), \qquad \gamma(m) = \frac{v_R}{v_R + A/m} = \frac{m}{m + m_\times}, \tag{5}$$

so $m_\times$ is the count at which shrinkage gives the direct estimate half its weight. Doubling m through n at fixed ρ, k and transmission law reduces process-mean risk by the fraction $1/[2(1 + m/m_\times)]$, which is one quarter at $m = m_\times$ and less beyond it. Changing ρ changes A as well as m. The scale $m_\times$ describes the selected estimator and target; it does not mark a point where information stops accumulating.

Corollary 1 (A fixed set of observed periods)**.** *Under H1–H4 and conditional independence of counts given $R_{1:T}$, let $\overline{R}_T$ denote the mean of $R_1, \ldots, R_T$ and $\overline{\hat{R}}_T$ the mean of $\hat{R}_1, \ldots, \hat{R}_T$. If the R_t share common marginal moments, the two mean targets have risks:*

$$\mathrm{E}[(\overline{\hat{R}}_T - \overline{R}_T)^2] = \frac{A}{Tm}, \qquad \mathrm{E}[(\overline{\hat{R}}_T - \overline{R})^2] = B_T + \frac{A}{Tm}, \qquad B_T = \mathrm{Var}(\overline{R}_T). \tag{6}$$

In Eq. (6)*, $B_T = v_R/T$ for independent, identically distributed R_t. For a stationary process with $\mathrm{Cov}(R_t, R_{t+h}) = v_R\phi^{|h|}$, $|\phi| < 1$, $B_T = v_R S(\phi, T)/T^2$ exactly, where the window sum is:*

$$S(\phi, T) = T + 2\sum\nolimits_{h=1}^{T-1}(T - h)\phi^h = \frac{T(1+\phi)}{1-\phi} - \frac{2\phi(1-\phi^T)}{(1-\phi)^2}. \tag{7}$$

The realised-mean identity needs only conditionally independent counting errors, not independent transmission levels. More cases in the same periods remove its counting error, while process-mean risk tends to B_T; only more periods reduce B_T, at a rate set by their dependence. Positive persistence reduces the gain from adjacent periods. Spatial averaging behaves analogously. For J units with transmission variance v_R and mean pairwise correlation $\bar{c}$, the averaged transmission level has variance

$$\mathrm{Var}(J^{-1}\textstyle\sum_{j=1}^{J} R_{j,t}) = v_R[\bar{c} + (1 - \bar{c})/J]. \tag{8}$$

Figure 1 plots these risks, with known-truth Monte Carlo checks, and Table 2 collects them.

Table 2 Inferential targets and exact mean squared errors. The fixed-reporting column assumes H1–H4; the volatile-reporting column assumes H2–H5 and, for averages, independent reporting probabilities independent of the transmission path. Counting errors are conditionally independent across periods. $B_T = \mathrm{Var}(\overline{R}_T)$ equals v_R/T for independent, identically distributed transmission and $v_R S(\phi, T)/T^2$ under the stated stationary autocovariance. With the previous observed count as denominator, the current-value entry under fixed reporting becomes approximately $[A + (1-\rho)(\overline{R}^2 + v_R)]/m$ (Eq. 9; Table 3).

Target	Estimator	Fixed, known reporting	Unobserved volatile reporting
Current R_t	$\hat{R}_t$	A/m	$v_H + A_H/m$
Realised mean $\overline{R}_T$	$\hat{\overline{R}}_T$	$A/(Tm)$	$(v_H + A_H/m)/T$
Process mean $\overline{R}$	$\hat{\overline{R}}_T$	$B_T + A/(Tm)$	$B_T + (v_H + A_H/m)/T$

Table 3 Point-target error with a known versus an observed denominator

m	Known denominator A/m	MC (SE)	Observed denominator, delta	MC (SE)	MC observed / (A/m)	Zero denominators	$+0.5$ variant MC (SE)
10	0.21232	0.21304 (0.00055)	0.31451	0.51453 (0.00482)	2.423	0	0.42529 (0.00276)
50	0.04246	0.04234 (0.00010)	0.06290	0.06745 (0.00019)	1.588	0	0.06589 (0.00018)
200	0.01062	0.01060 (0.00002)	0.01573	0.01601 (0.00004)	1.508	0	0.01592 (0.00004)
1000	0.00212	0.00212 (0.00000)	0.00315	0.00316 (0.00001)	1.489	0	0.00316 (0.00001)

Parameters $\bar{R} = 1.15$, $v_R = 0.04$, $k = 0.35$, $\rho = 0.25$ ($A = 2.1232$); 400,000 Monte Carlo draws per m. The delta column is $[A + (1-\rho)(\bar{R}^2 + v_R)]/m$. Crossing scale $m_\times = A/v_R = 53.08$ with a known denominator and $[A + (1-\rho)(\bar{R}^2 + v_R)]/v_R = 78.63$ when the denominator is the observed primary-generation count (delta approximation). At $m = 10$ the delta approximation understates the Monte Carlo error of the observed-denominator estimator. Draws with a zero primary-generation count are excluded from the ratio estimator (share shown); the $+0.5$ variant $(C_t + 0.5)/(M_t + 0.5)$ keeps every draw.

2.3 An observed denominator

Surveillance ratios divide by an observed count rather than by a known scale. Let the denominator be the reported count of the primary generation, $M_t \sim \mathrm{Bin}(n, \rho)$, independent of the offspring given n, and let $\tilde{R}_t = C_t/M_t$. Because $M_t = m(1 + u_t)$ with $\mathrm{Var}(u_t) = (1-\rho)/m$, a delta expansion (Appendix A.5) gives the approximate risk:

$$\mathrm{E}[(\tilde{R}_t - R_t)^2] \approx \frac{A + (1-\rho)(\overline{R}^2 + v_R)}{m}, \tag{9}$$

so the crossover scale moves to $m_\times + (1-\rho)(\overline{R}^2 + v_R)/v_R$. With $\overline{R} = 1.15$, $v_R = 0.04$, $k = 0.35$ and $\rho = 0.25$ ($A = 2.1232$) it rises from 53.1 to 78.6. Monte Carlo with 400,000 draws per scale (Table 3) puts the observed-denominator error at 1.59, 1.51 and 1.49 times A/m for $m = 50$, 200 and 1000, close to the delta value of 1.48. At $m = 10$ the ratio is 2.42 and the delta form understates the error, because small denominators inflate the ratio. The ratio estimator is undefined when $M_t = 0$; such draws are excluded and their share is reported in Table 3, where it is zero to the precision shown at every scale considered. At $m = 10$ the Monte Carlo error is 0.5145 with standard error 0.0048. A continuity-corrected ratio, $(C_t + 0.5)/(M_t + 0.5)$, lowers it to 0.4253, still 2.00 times A/m.

3 Reporting volatility and identification

3.1 Period-level reporting variation

Replace H1 by a common period-specific reporting probability $p_t \in [0, 1]$, independent of R_t and of n, with known mean $\rho > 0$ and $\mathrm{Var}(p_t) = \rho^2\tau^2$ (H5). Reporting remains independent across infections given p_t. Because p_t lies in $[0, 1]$ with mean ρ, $\mathrm{Var}(p_t) \le \rho(1-\rho)$ [35], so $\tau^2 \le (1-\rho)/\rho$. The estimator still divides by $m = \rho n$ and does not observe p_t.

Proposition 2 (Reporting volatility and second-moment nonidentification)**.** *Under H2–H5, put $Q = \overline{R}^2 + v_R$. The first two moments of $\hat{R}_t$ are:*

$$\mathrm{E}(\hat{R}_t) = \overline{R}, \qquad \mathrm{Var}(\hat{R}_t) = v_R + v_H + \frac{A_H}{m}, \qquad v_H = \tau^2 Q, \qquad A_H = \overline{R} + \frac{\rho(1+\tau^2)Q}{k}. \tag{10}$$

For known m, ρ and k, the single-period mean and variance in Eq. (10) *depend on v_R and τ^2 only through $U = (1 + \tau^2)(\overline{R}^2 + v_R)$.*

Given R_t and p_t the estimator has mean $(p_t/\rho)R_t$, so a shared reporting shock is not averaged away by more infections in the same period. The statement is deliberately narrow. It concerns the first two moments of one period's count within the specified family. It does not exclude identification from full count distributions, from temporal structure under further assumptions, or from external measurements of reporting; it also assumes that reporting is independent of transmission, an assumption Section 6 relaxes.

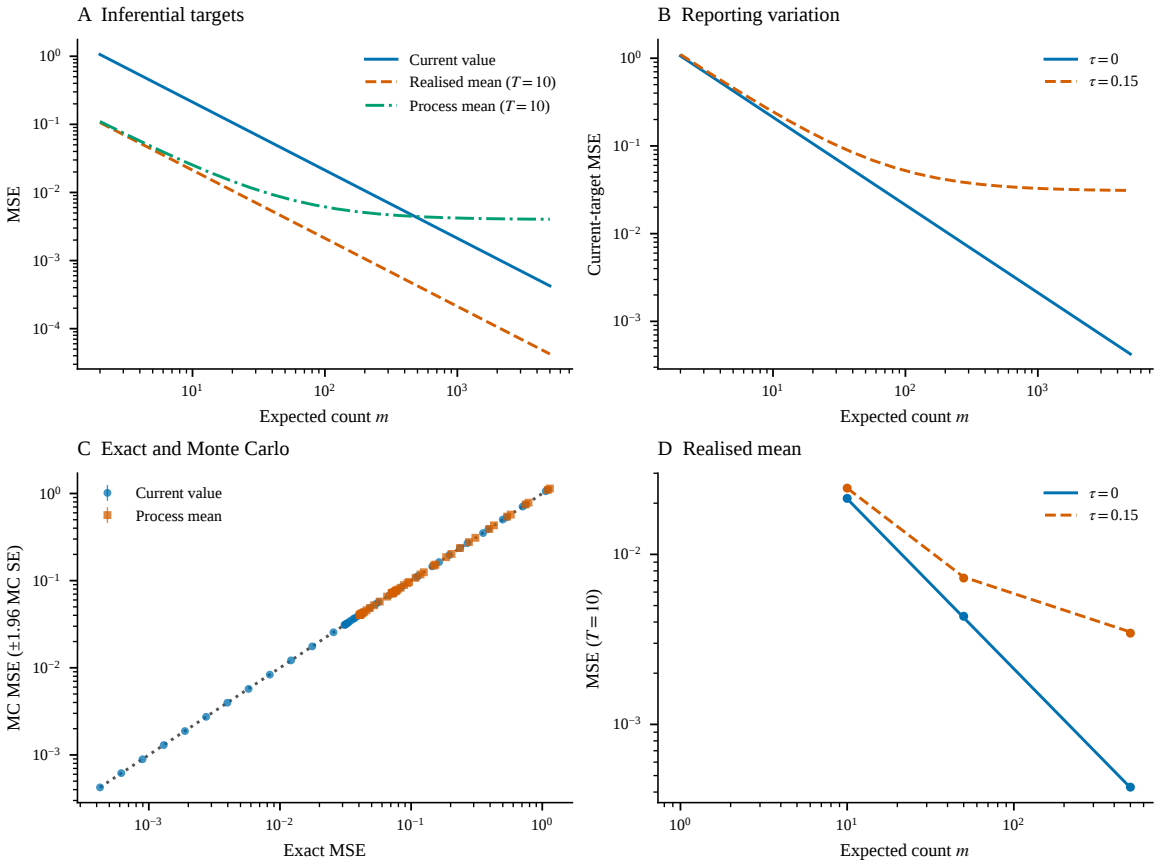


Figure 1 Model risks for the three targets. Panel A shows the mean squared error against expected count m for the current value, the realised mean and the process mean over $T = 10$ independent periods. Panel B shows the current-value risk with fixed reporting ($\tau = 0$) and with period-level reporting variation ($\tau = 0.15$). Panel C shows the exact risks against known-truth Monte Carlo estimates with 1.96 Monte Carlo standard errors. Panel D shows the realised-mean risk over $T = 10$ periods at three scales, with $\overline{R} = 1.15$, $v_R = 0.04$, $k = 0.35$ and $\rho = 0.25$. Counts are simulated under the fixed-exposure model, so the curves describe that model and not the hospital data.

Corollary 2 (Current and realised-mean risks with volatile reporting)**.** *Under H2–H5, the current-value risk is:*

$$\mathrm{E}[(\hat{R}_t - R_t)^2] = v_H + \frac{A_H}{m}. \tag{11}$$

If, in addition, the p_t are independent and identically distributed, independent of the transmission trajectory, and counts are independent given both trajectories, the two mean targets have risks:

$$\mathrm{E}[(\overline{\hat{R}}_T - \overline{R}_T)^2] = \frac{v_H + A_H/m}{T}, \qquad \mathrm{E}[(\overline{\hat{R}}_T - \overline{R})^2] = B_T + \frac{v_H + A_H/m}{T}. \tag{12}$$

At fixed T, unobserved reporting volatility leaves the floor v_H/T on the realised-mean target as m grows, so a target free of transmission variation is not free of reporting error. The crossover scale with volatile reporting, $m_{\times,H} = A_H/(v_R+v_H)$, is never larger than $m_\times$ (Eq. (A.5)); the lower crossover accompanies a larger risk, not better surveillance. Proofs are in Appendix A.4.

3.2 From the branching experiment to surveillance counts

If the scale m is random and associated with transmission, conditional unbiasedness given (R_t, m) still yields the conditional risk:

$$\mathrm{E}[(\hat{R}_t - \overline{R})^2 \mid m] = \frac{\mathrm{E}(R_t \mid m) + \rho\,\mathrm{E}(R_t^2 \mid m)/k}{m} + \mathrm{Var}(R_t \mid m) + [\mathrm{E}(R_t \mid m) - \overline{R}]^2, \tag{13}$$

and the conditional moments in Eq. (13) can change with scale. A constant intercept plus a constant coefficient on inverse count therefore does not follow when the count scale is itself an epidemic outcome, as it is in surveillance. Delays, aggregation, ratio denominators and changing conditional means further separate an apparent statistic from the experiment above. The rest of the paper accordingly treats empirical dispersion diagnostics as descriptive quantities and asks, with a known-truth audit, what they recover.

4 What cross-sectional dispersion diagnostics measure

4.1 Data and apparent statistics

We used the National Healthcare Safety Network (NHSN) weekly hospital respiratory-data snapshot retrieved on 20 October 2026 and restricted influenza admissions to the 50 US states and the District of Columbia [36, 37]. Hospital participation varied during the series, including a voluntary-reporting interval from May to October 2024 before reporting became mandatory again [38]. The analysis describes this surveillance system and does not estimate an effect of the policy change.

The windows were the early-growth phases of four seasons: 1 October–26 November 2022, 7 October–23 December 2023, 9 November–28 December 2024 and 4 October–27 December 2025. Each is the contiguous national segment with weekly growth of at least 1.05 before the first turning point after the season's admissible start; the 2024 start follows the return to mandatory reporting. The rule is a fixed definition applied identically to every season, and it was not varied in the primary analysis. Inference conditions on the selected windows, and Table 12 refits the model with growth thresholds of 1.02 and 1.10. Missing or suppressed counts were kept missing rather than set to zero. A row requires four observed counts on consecutive weekly dates, from two weeks before its denominator week through the following week.

With C_t the admissions in week t, we used two apparent statistics. The first is the weighted growth ratio:

$$G_t = \frac{C_{t+1}}{D_t}, \qquad D_t = 0.65C_t + 0.25C_{t-1} + 0.10C_{t-2}, \tag{14}$$

The second is the growth-to-reproduction transform:

$$R_{A,t} = [\max\{0,\, 1 + (\theta/7)\log\{(C_{t+1} + 0.5)/(C_t + 0.5)\}\}]^{\alpha}, \qquad \alpha = (g_\mu/g_\sigma)^2, \quad \theta = g_\sigma^2/g_\mu, \tag{15}$$

The transform follows the renewal relation between growth rate and reproduction number for a gamma-distributed interval [39]. Neither statistic is a true reproduction number or a posterior estimate from a renewal model [3, 10]. Rows require $D_t \geq 5$ and, for R_A, $C_t \geq 5$, which leaves 1,748 state-weeks for R_A and 1,756 for G.

The primary interval has mean $g_\mu = 3.6$ days and standard deviation $g_\sigma = 1.6$ days, the household serial-interval estimate of Cowling et al. [40]. It was kept as the primary choice because it is directly estimated from influenza household transmission and is widely used in renewal-based influenza work, even though a serial interval is not a generation interval [41]. Two sensitivities use a mean of 3.2 days, either with the same coefficient of variation or with standard deviation 2.1 days; the latter is the intrinsic generation-time estimate from US household data [42]. Section 5.3 shows how much the choice matters.

4.2 The diagnostic and its three summaries

For an apparent statistic X with pooled mean $\bar{X}$, define the relative squared deviation and its affine working model

$$L_{st} = \frac{(X_{st} - \bar{X})^2}{\bar{X}^2}, \qquad \mathrm{E}(L_{st} \mid C_{st}) = a + \frac{b}{C_{st}}, \tag{16}$$

For G, D_t of Eq. (14) replaces C_t. This is the empirical counterpart of $V(m)/\overline{R}^2$ in Eq. (3), but with a realised count in place of the expected scale and an apparent statistic in place of $\hat{R}_t$. Three summaries of the fit of Eq. (16) follow.

The intercept a is the count-independent relative dispersion. It measures the sum of all relative variation in the diagnostic statistic that does not scale with count. When reporting probabilities do not vary from week to week ($\tau = 0$), a estimates the relative variance of transmission, $\tilde{v}_R = v_R/\overline{R}^2$. When reporting shocks vary between weeks, a estimates $\tilde{v}_R + \tilde{v}_H$, where $\tilde{v}_H$ is the relative variance that the shocks induce in the ratio statistic; a reporting level that is constant over consecutive weeks cancels in the ratio (Appendix A.6). By Proposition 2, single-period cross-sectional moments cannot separate the two components. The intercept is therefore not an estimate of an error floor. With $\tau = 0$ the current-value error floor is zero, and with $\tau > 0$ the floor $\tilde{v}_H$ does not exceed a, which is only an upper bound for it. Below we call a the count-independent relative dispersion, or the relative dispersion for short.

The counting slope b is read against two references. A delta expansion of Eq. (15) under independent Poisson counts in consecutive weeks gives the independent-Poisson reference $b_{\mathrm{P}} = \overline{\kappa(g)^2[1 + (C_t + 0.5)/(C_{t+1} + 0.5)]}$, a weighted mean over rows, with $\kappa(g) = (g_\mu/7)/[1 + \theta\log(g)/7]$ and $g = (C_{t+1} + 0.5)/(C_t + 0.5)$. For the primary sample $b_{\mathrm{P}} = 0.4455$. The term 1 is the Poisson variance of the denominator count. In a branching chain the denominator week contains the parents of the numerator week, so conditioning on the realised parents removes the transmission part of that variance and leaves only the binomial reporting part, $(1 - \rho)/C_t$. With Poisson offspring and reporting probability ρ this gives the branching-Poisson reference and its ratio to b_{P}:

$$b_{\mathrm{P,br}}(\rho) = \overline{\kappa(g)^2[(1 - \rho) + (C_t + 0.5)/(C_{t+1} + 0.5)]}, \qquad r(\rho) = \frac{b_{\mathrm{P,br}}(\rho)}{b_{\mathrm{P}}} \approx \frac{1 + (1 - \rho)g}{1 + g} = 1 - \frac{\rho g}{1 + g}, \tag{17}$$

Table 4 Mapping from theoretical inferential targets to empirical surveillance diagnostics

Theoretical Target (Section 2)	Empirical Diagnostic (Section 4)	Exact Equivalence Conditions	Distortion Mechanism and Bias Direction
Relative transmission variance: $\tilde{v}_R = v_R/\overline{R}^2$	Dispersion Intercept: a	Constant reporting probability ($\tau = 0$); scale-independent transmission ($\gamma = 0$).	If reporting varies ($\tau > 0$), $a \approx \tilde{v}_R + \tilde{v}_H$. The intercept is an upper bound on $\tilde{v}_R$, not an error floor. Under scale dependence ($\gamma > 0$), $a < \tilde{v}_R$.
Counting error scale: $\tilde{A} = A/\overline{R}^2$	Counting Slope: b	Independent Poisson limit ($\rho \to 0$ gives $b \approx b_{\mathrm{P}}$); branching Poisson ($b \approx b_{\mathrm{P,br}}(\rho)$).	Inflated by offspring overdispersion ($k < \infty$, $b > b_{\mathrm{P,br}}$) or unmodelled scale-dependent heterogeneity ($\gamma > 0$).
Crossover scale: $m_\times = A/v_R$	Crossover Count: b/a	Exact match between A and b, and between v_R and a.	Under reporting shocks ($\tau > 0$), b/a is attenuated below $m_\times$; under scale dependence ($\gamma > 0$), b/a is inflated. b/a is an in-sample balance point.

Formal correspondence between the theoretical parameters derived under the branching model in Section 2 and the empirical summaries estimated by the cross-sectional dispersion diagnostic in Section 4. The table makes explicit the conditions under which empirical diagnostics identify theoretical quantities and the sign of the distortion when those conditions fail.

The row-level ratio $r(\rho)$ falls to $1/(1+g) \approx 0.42$ at $\rho = 1$ for the geometric-mean growth $g_0 = 1.407$. Averaged over the observed rows, $r(\rho)$ is 0.944, 0.861, 0.722 and 0.443 for $\rho = 0.1$, 0.25, 0.5 and 1 (Table S4). The two references correspond to the two estimators of Section 2. The observed-denominator model of Section 2.3, whose denominator noise $(1-\rho)/m$ is the reporting part only, has the structure of the branching-Poisson reference, and the known-scale estimator of Section 2.1 is its limit without denominator noise. The independent-Poisson reference is the limit $\rho \to 0$, in which the denominator is an independent Poisson count. The ratio b/b_{P} therefore asks whether counting variation exceeds independent-Poisson noise, and $b/b_{\mathrm{P,br}}(\rho)$ asks the same question for a branching chain, at a reporting probability that the data do not identify.

The third summary is the dispersion-to-counting crossover count b/a, the count at which the fitted counting term equals the relative dispersion and the empirical analogue of $m_\times$. It is defined on the apparent statistic's count scale, in admissions per week, and differs from $m_\times$, which refers to the expected count of a branching experiment. Table 4 synthesises the formal correspondence between the theoretical parameters $(\tilde{v}_R, \tilde{A}, m_\times)$ derived in Section 2 and the empirical summaries $(a, b, b/a)$ defined here, detailing the conditions under which empirical diagnostics identify theoretical quantities and the sign of distortion when those conditions fail.

The affine mean was fitted by iterated feasible generalised least squares with weights inversely proportional to the squared fitted mean, which minimises a gamma quasi-deviance; this is an estimating criterion, not a distributional claim. Intervals are percentiles of state-by-week pigeonhole bootstrap draws in which each row receives the product of independently drawn state and week multiplicities [43], a resampling counterpart of two-way cluster-robust inference [44]; the pooled mean and the regression are re-estimated in every draw. The primary bootstrap has 2,000 draws, and every pooled row in every table uses its weights. The within-state and two-way slopes, the season models, the generation-interval sensitivity and the sensitivity specifications also use 2,000 draws. The two- and three-week block versions of the fixed-effects decomposition, which check sensitivity to persistent national shocks, use 1,000 draws each, as do the thinning analysis of the data and the held-out validation. Draws whose iterated fit does not converge are dropped, and their number is reported with each table.

The audit below shows that these bootstrap intervals are conservative, so a second interval was calibrated for the headline summaries. It is the two-way cluster-robust sandwich of Cameron et al. [44] for the stacked estimating equations of the pooled mean and the affine fit, clustered by state and by calendar week. Small-sample factors $G/(G-1)$ are applied, critical values come from the t distribution on 41 degrees of freedom, one fewer than the 42 weeks, and the interval for b/a is formed on the log scale. A variance-ratio-scaled bootstrap, in which the percentile interval is shrunk towards the point estimate by the bootstrap-to-Monte-Carlo standard deviation ratio of the other coverage generators, was the alternative. The rule for choosing between them was fixed before the coverage runs: the sandwich is used if its coverage of the estimator mean is at least 0.925 under every coverage generator, and the scaled bootstrap otherwise. The same sandwich gives the covariance of the season intercepts, and the season test is referred to a simulated null distribution (Section 4.4).

Binomial thinning provides a fourth check. Each unique state-week count is thinned once to $\mathrm{Bin}(C, q)$ for $q \in \{0.8, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1\}$, and the thinned value is reused in every ratio that contains it. Recomputing the statistic and the fit, with the original C_t as regressor, gives the thinning response:

$$b(q) - b(1) \simeq c\,(1/q - 1). \tag{18}$$

Under the delta approximation $c = b_{\mathrm{P}}$ whatever the original counting variation, because thinning adds known binomial variance. The ratio c/b_{P} is therefore a check of that approximation and not of the data-generating process (Appendix B.4).

4.3 Known-truth audit on the real design

The audit is a simulation study with known truth and is reported under the aims, data-generating mechanisms, estimands, methods and performance measures structure of Morris et al. [45]. Its aim is to establish what a, b, b/a and c recover on the sampling design of the application. Synthetic counts were generated on the exact R_A design: the 1,748 rows, 1,969 unique state-weeks and 210 state runs of the data. Each run is simulated as a whole with shared counts, so that one week's count is the next week's denominator. The expected count is carried forward by the true growth factors, and the geometric-mean observed growth $g_0 = 1.407$ is used for transitions that are not scored. The simulated C_t serves as the count scale, the filter $C_t \geq 5$ is re-applied, and the estimand of each row is the transform (15) of its true growth factor.

Two alignment steps fix the level of the synthetic panels, and the quantities they fix are calibration targets rather than tests. Each run is level-anchored to the geometric mean of its observed counts, and one scale factor per generator, between 0.797 and 0.844, sets the Monte Carlo mean of the median retained count to the observed median of 45. In every generator with transmission heterogeneity, the amplitude of that heterogeneity is then set so that the Monte Carlo mean of the fitted intercept equals the observed a. The slope ratio b/b_{P}, the crossover b/a, the thinning rate, the ratio of pooled to within-state slope, the reduction of a under week centring and the score contrast between affine and power-law fits are not targeted and serve as tests. A generator passes a test when the observed point value lies inside its 2.5–97.5% Monte Carlo range, and passes the score contrast when the observed contrast lies between the 0.025 and 0.975 quantiles of its simulated contrasts. This decision rule and the week-centring test were fixed after the generators had first been compared with the data, and Table 9 applies them to every generator. The lower tail of the retained counts matches the data (10th percentiles 10–11 against 10). The upper tail does not, since the generators' 90th percentiles are 263–324 against 389 observed (Table 7). Post-stratifying retained rows to the observed deciles of the count changes intercept recovery from 1.007 to 1.008 under S1, from 1.115 to 1.079 under S4, from 1.009 to 1.004 under S6 and from 0.684 to 0.665 under S7a, so the truncated tail moves recovery by at most 3.2%. Because the implied transmission relative standard deviation scales with the inverse square root of the recovery ratio, the reweighting moves it by at most 1.7%, far less than the width of the ranges reported below.

Table 5 sets out the design. The eleven generators span the mechanisms that the diagnostics are meant to separate, from independent Poisson counts (S0) to scale-dependent heterogeneity combined with state-week reporting shocks (S8). A twelfth generator, S9, is S8 with half of the transmission heterogeneity carried by the national calendar-week component of S4 and S5. It was added after the first eleven had been compared with the data, and neither its share nor its week factor was tuned to them.

Each generator was run on 200 panels, so the quantile of an observed value among simulated values has a resolution of 0.005; the score-contrast quantiles pool five independent streams of 200 panels and are reported with binomial Monte Carlo standard errors. Thinning used 100 panels with 20 draws per level. Every panel passes through the same filters and fitting functions as the data. The audit covers R_A; the G statistic was not audited.

Two further designs address the questions that the generator list leaves open. A grid over the share ϕ of the total relative variance carried by transmission, $\phi \in \{0, 0.25, 0.5, 0.75, 1\}$, crossed with scale dependence $\gamma \in \{0, 0.25, 0.5\}$, uses 1,000 panels per cell, with the remainder $1 - \phi$ carried by state-week reporting shocks. In its calibrated rows the amplitude makes the mean intercept equal the observed one; an uncalibrated block at $\gamma = 0.25$ fixes the total variance at the observed a and tests a as well. A coverage study analyses 400 panels from each of S1, S6, S8 and S9 exactly as the data, with 200 pigeonhole bootstrap draws per panel, and records whether each nominal 95% interval covers its target. The same 400 panels per generator calibrate the two-way sandwich and the season test; under the season null each season intercept is centred at its Monte Carlo mean.

Two designs complete the tests. A second share grid crosses ϕ and γ with a national calendar-week share of the transmission draw of 0 or 0.5, giving 30 cells, and applies all six tests to each cell with the decision rule of the audit. In each cell with heterogeneity the amplitude is also recalibrated to the lower and upper limits of the calibrated interval for a, so that the transmission relative standard deviation implied by the observed intercept carries the uncertainty in a. A held-out reference repeats the leave-one-state-out and leave-one-season-out score comparison of Section 5.4 on 200 synthetic panels from every generator, with the folds, rows and score of the data.

The estimand of the audit is operational. The intercept a is compared with the relative variance of the transmission factor on the retained rows of each simulated panel, $\tilde{v}_R$, and, when reporting shocks are present, with the total $\tilde{v}_R + \tilde{v}_H$. These are finite-population quantities of the panel, not parameters of a superpopulation of states or seasons [27]. Interval coverage is judged against the Monte Carlo mean of each estimator, which a dependence-aware resampling scheme can target, and separately against the true total. The targets of Section 2 map onto these quantities only under the conditions of Section 3. With reporting fixed, $\tilde{v}_R$ is the between-row variation that a current-value analysis resolves; with volatile reporting the total bounds the error floor of the current value.

4.4 Audit results

Without reporting shocks the intercept tracks the relative variance of transmission. When transmission is constant (S0, S2 and S2P) the fitted intercept is essentially zero, so b/a is undefined. With independent heterogeneity, the ratio of a to the true relative variance of R_i on the retained rows is 1.01 (Monte Carlo range 0.93–1.09) under S1 and 1.02 (0.91–1.13)

Table 5 Design of the known-truth audit

Element	Specification
Aim	What a, b, b/a, b/b_{P} and the thinning rate c recover on the R_A design of the application
Estimands	Transmission relative variance $\tilde{v}_R$; total $\tilde{v}_R + \tilde{v}_H$ when reporting shocks are present; expected slope relative to b_{P}
Methods	The pipeline of the application: R_A transform, relative squared deviation, gamma FGLS affine fit, plug-in b_{P}, within-state slope, week centring, score contrast between power-law and affine fits (positive values favour the affine form)
Performance measures	Monte Carlo mean with its standard error; 2.5% and 97.5% Monte Carlo limits; ratio of means for recovery; quantile of the observed value among simulated values with binomial standard error; interval coverage
Calibration targets	Median retained count (all generators); mean fitted intercept equal to the observed a (generators with heterogeneity)
Tests	b/b_{P}, b/a, thinning rate, ratio of pooled to within-state slope, week-centring reduction of a, score contrast; a test is passed when the observed value lies inside the 2.5–97.5% Monte Carlo range (contrast: quantile in $[0.025, 0.975]$)
Replicates	200 panels per generator; five streams of 200 for score-contrast quantiles; thinning 100 panels with 20 draws per level; share grid five streams of 200 per cell; $\rho \times k$ grid 100 panels per cell; coverage 400 panels per generator (S1, S6, S8, S9) with 200 bootstrap draws each; interval and season-test calibration 400 panels per coverage generator (bootstrap-covariance season test 200 panels with 200 draws); six-test share grid 30 cells with 200 panels, 1,000 contrast panels and thinning 100 panels with 20 draws per cell; held-out reference 200 panels per generator
Data-generating mechanisms	
S0	Constant growth g_0, independent Poisson counts
S1	Independent lognormal growth factors ($\sigma = 0.247$), Poisson counts
S2	Constant growth, negative binomial offspring ($k = 0.3$), binomial reporting at $\rho = 0.25$
S2P	S2 with Poisson offspring (pure branching chain with reporting)
S3	S1 combined with the S2 counting model
S4	Growth heterogeneity split equally between state and national-week components, Poisson counts
S5	S4 combined with the S2 counting model
S6	S3 with a state-week reporting shock $\rho_{st} = 0.25 z_{st}$, $\log z_{st} \sim N(-s^2/2, s^2)$, $s = 0.122$
S7a, S7b	S3 with heterogeneity $\sigma^2(\mu) = \sigma_0^2(\mu/45)^{-\gamma}$ for expected count μ, $\gamma = 0.25$ and 0.5
S8	S3 with scale-dependent heterogeneity ($\gamma = 0.25$) and state-week reporting shocks ($s = 0.133$)
S9	S8 with half of the transmission heterogeneity carried by the national-week component of S4 ($s = 0.132$)

In S6, S8 and S9 the shock scale is chosen so that transmission and the shock each carry about half of the variance target set by the observed a, the $\phi = 0.5$ cell of the share grid. Tests are statistics that no alignment step targets.

under S3. With structured heterogeneity it is 1.12 (1.04–1.19) under S4 and 1.15 (1.06–1.27) under S5 (Table 7; Fig. 2C), because crossed state and national-week components partially project onto the marginal intercept. Fitting the same affine model to the true estimation error, $((X - R_i)/\bar{X})^2$, gives an intercept of about 3×10^{-5} in each of these generators, so the current-value error floor is essentially zero without reporting shocks. Week centring removes 62% of a under S4 and 59% under S5, where half of the heterogeneity is national, and essentially none under S1 and S3.

With reporting shocks the intercept is the sum of both components. Under S6, a is 1.01 (0.87–1.14) times the total $\tilde{v}_R + \tilde{v}_H$ and 2.02 (1.71–2.29) times $\tilde{v}_R$ alone. The true-error intercept is 1.02 (0.83–1.19) times $\tilde{v}_H$, so the current-value error floor equals the shock component and is about half of a. The shock terms of consecutive rows have correlation -0.50 (-0.53 to -0.46), because a shock in one week enters the numerator of one row and the denominator of the next. Scale-dependent heterogeneity biases the intercept in the other direction. Under S7a and S7b it is 0.68 (0.60–0.78) and 0.43 (0.37–0.50) times the true relative variance, because variation that shrinks with count is absorbed by the slope. Under S8 the two distortions combine, and a is 1.75 (1.47–2.04) times $\tilde{v}_R$ and 0.85 (0.72–0.99) times the total.

The counting slope does not identify the mechanism. The ratio b/b_{P} is 0.93 (0.85–1.01) under S0, 0.80 (0.73–0.86) under S2P and 1.23 (1.13–1.32) under S2 (Table 8; Fig. 2A). The gamma quasi-likelihood slope therefore falls about 7% short of the analytic reference under independent Poisson counts, and the simulated-to-analytic ratio is 0.91–0.93 for every reference in Table S4. The step from S0 to S2P reflects the structure of denominator noise. In a branching chain only the reporting part of the denominator noise remains, as the branching-Poisson reference predicts ($r(0.25) = 0.861$, simulated 0.802). The step from S2P to S2 is offspring overdispersion. Transmission heterogeneity moves the ratio in both directions. It is 0.85 (0.69–1.02) under S1, 1.14 (0.98–1.31) under S3, 0.73 (0.60–0.87) under S4, 1.03 (0.87–1.20) under S5 and 1.14 (0.97–1.34) under S6. Scale-dependent heterogeneity raises it to 1.56 (1.34–1.79) under S7a, 2.25 (1.98–2.58) under S7b and 1.31 (1.08–1.52) under S8. A slope near or above b_{P} is therefore compatible with several combinations of overdispersion, reporting and heterogeneity. Structured heterogeneity also lowers the within-state slope below the pooled slope, although counting noise is the same in every state. Under S4 the ratio of pooled to within-state slope is 1.14 (0.99–1.37), and in the other generators it lies between 1.01 and 1.10. The observed point ratio, 1.48, lies

outside the Monte Carlo range of every generator, although its interval, 0.99–3.66, reaches 1.

The crossover count does not discriminate between mechanisms either. Among generators with heterogeneity, b/a ranges from 22.2 (17.0–27.7) under S4 to 42.1 (30.8–58.2) under S8, against 51.4 (39.5–65.9) and 77.5 (58.4–99.7) under S7a and S7b. The observed value, 37.4, lies within the S3, S5, S6 and S8 ranges, and both its bootstrap interval, 18.4–72.4, and its calibrated two-way interval, 24.1–58.3, overlap all eight.

The curve shape separates generators more sharply than either slope summary. The in-sample difference in quasi-deviance score between affine and power-law fits is close to zero under S0, S2 and S2P, whose intercept is zero. Generators with heterogeneity that does not depend on count give differences from 0.012 (S1) to 0.024 (S5) in favour of the affine form, whereas the observed difference is -0.0014. At most 0.3% of their panels reach a contrast this low (Fig. 2D). Under S7a the observed contrast lies at the 0.495 quantile, but the S7a slope ratio range (1.34–1.79) excludes the observed 1.257. Under S8 the observed contrast lies at the 0.069 quantile (Monte Carlo standard error 0.008), and the observed crossover and slope ratio lie inside the S8 ranges. S8 does not reproduce the two remaining tests. Its pooled-to-within slope ratio ranges from 0.92 to 1.15 against the observed 1.48, and week centring changes its a by -6.1% to 2.3% against the observed reduction of 34.7%. S9, which adds a national calendar-week component to S8, raises the week-centring reduction to 19.1% (12.5–25.9), still below the observed value, and leaves the slope-ratio range at 0.92–1.17; its observed contrast quantile is 0.057 (0.007). No generator passes every test, and S8 and S9 pass four of six (Table 9). The contrast comparison is in sample, but its held-out version points the same way (Table S18). Leaving one state out, the observed power-minus-affine contrast, -0.0020, lies below every simulated panel under S1, S3, S4, S5 and S6, whose mean held-out contrasts are 0.011–0.024 in favour of the affine form, at the 0.125–0.475 quantiles of S0, S2, S2P and S7a, and at the 0.995 quantile of S7b. Under S8 and S9 it lies at the 0.050 (0.015) and 0.055 (0.016) quantiles, inside the decision rule but in the lower tail. With only four folds, the leave-one-season-out contrast discriminates weakly: its observed value, 0.0046, lies inside the central 95% of S0, S1, S6, S7a, S8 and S9 and outside it for the other six generators. The near-zero held-out state contrast, which on its own does not separate the two forms, is therefore unlikely under heterogeneity that does not depend on count. The simulated panels also retain 1,596 (S8) and 1,571 (S9) rows on average against 1,748 observed, and their upper count tail is shorter than the observed one. The audited generators therefore bracket the data on most summaries without reproducing all of them, and the data, which do not trace infections longitudinally, cannot identify the data-generating process.

The share grid shows how much the compatible region leaves open (Table S1; Fig. 3). Among calibrated cells, the observed slope ratio, crossover and contrast all fall inside the simulated ranges only at $\gamma = 0.25$ with $\phi = 0.25$, 0.5 and 0.75, and at $\gamma = 0.5$ with $\phi = 0.5$. Across these cells the intercept is 1.04–3.85 times the transmission relative variance and 0.74–0.94 times the total. Two of the compatible cells are borderline on the contrast, with quantiles of 0.026 and 0.024 (standard error 0.005). Without scale dependence ($\gamma = 0$) every share is excluded by the contrast, and with scale dependence pure transmission ($\phi = 1$) is excluded by the slope ratio and the crossover. Within the compatible region the transmission share is not identified. In the uncalibrated block, where the total variance is fixed at the observed a, only the cells with $\phi \leq 0.25$ keep a inside its range, and the contrast excludes both.

Under all six tests no cell of the extended grid reproduces the data (Table S2). The pooled/within slope ratio fails in all 30 cells: the largest simulated upper limit is 1.25 against the observed 1.48. Week centring passes in three cells, all with a national share of 0.5, and the contrast in 12. The best cell, $\gamma = 0.25$, $\phi = 0.75$ with a national share of 0.5, passes five of the six tests, and its implied transmission relative standard deviation is 11.5% (9.9–13.0% when the uncertainty in a is propagated). Nine cells pass the slope ratio, the crossover and the contrast, the tests of the first grid; thinning excludes none of them. Their implied relative standard deviation is 6.2–13.9% at the observed a and 5.3–15.5% with the uncertainty in a propagated, against 5.1–20.5% across all 25 cells with heterogeneity. Of the 180 pass or fail decisions, 19 lie within two Monte Carlo standard errors of a limit. Sensitivity analysis across decision rules confirms this structure: under a 90% rule, 8 of 180 decisions change (4.4%); under a 99% rule, 20 change (11.1%); and under a ± 2 MCSE tolerance band, 8 change (4.4%). Exactly 0 cells pass all six tests under any of these rules.

To resolve the conditionality of the implied transmission standard deviation on $k = 0.3$ and $\rho = 0.25$, Table 6 evaluates the entire (ρ, k) parameter manifold under $S1$ heterogeneity. Out of 20 parameter combinations, 14 yield counting slopes compatible with the empirical 95% confidence interval ($[0.81, 1.73]$). Across all 14 compatible cells, the recovery ratio a/v_R remains remarkably stable between 0.996 and 1.102. The corresponding implied transmission relative standard deviation spans 11.6–12.3% at the point estimate and 9.9–13.8% with the uncertainty in a propagated. Expanding across the full mechanism space of reporting shocks and scale dependence yields a global envelope of 5.0–20.4%. Furthermore, decomposition of the pooled slope ($b = 0.560$) reveals that it is a weighted combination of the within-state slope (0.379) and a positive between-state slope (0.680), driven by smaller jurisdictions exhibiting higher baseline dispersion. The 95% bootstrap interval of the pooled-to-within slope ratio is $[0.988, 3.662]$, which spans 1.0 and indicates that the within-between divergence is statistically compatible with sample variability.

The bootstrap intervals are conservative for their estimands (Table 10). Across S1, S6, S8 and S9, intervals for a, b and b/a cover the Monte Carlo mean of the corresponding estimator in at least 99.5% of panels, against a nominal 95%. The mean bootstrap standard deviation is 1.6–2.5 times the Monte Carlo standard deviation of the point estimates, in line with the conservative variance of the pigeonhole bootstrap under crossed designs [43]. Coverage of the true total relative variance by the interval for a is 1.000 under S1 and 0.998 under S6, but falls to 0.892 (0.015) under S8. That shortfall

Table 6 Counting-slope compatibility, recovery ratio, and implied transmission relative SD across the (ρ, k) parameter manifold

ρ	k	b/b_{P} [95% MC range]	Slope compatible	a/v_R	Implied SD point, %	Implied SD 95% CI, %
0.1	0.1	1.28 [1.11, 1.47]	✓	1.022	12.1	10.3–13.7
0.1	0.3	0.98 [0.86, 1.13]	✓	1.009	12.2	10.4–13.7
0.1	1	0.86 [0.69, 1.01]	✓	1.011	12.2	10.4–13.7
0.1	3	0.82 [0.69, 0.99]	✓	1.010	12.2	10.4–13.7
0.1	∞	0.81 [0.68, 0.94]	✓	0.998	12.2	10.4–13.8
0.25	0.1	1.80 [1.59, 2.03]	✓	1.059	11.9	10.1–13.4
0.25	0.3	1.14 [0.95, 1.31]	✓	1.019	12.1	10.3–13.7
0.25	1	0.86 [0.71, 1.02]	✓	1.006	12.2	10.4–13.8
0.25	3	0.79 [0.67, 0.93]	✓	0.999	12.2	10.4–13.8
0.25	∞	0.76 [0.61, 0.89]	✓	0.996	12.3	10.4–13.8
0.5	0.1	2.40 [2.07, 2.74]	–	1.170	11.3	9.6–12.8
0.5	0.3	1.41 [1.24, 1.59]	✓	1.032	12.0	10.3–13.6
0.5	1	0.85 [0.69, 0.98]	✓	1.025	12.1	10.3–13.6
0.5	3	0.71 [0.59, 0.81]	–	1.001	12.2	10.4–13.8
0.5	∞	0.62 [0.50, 0.75]	–	1.002	12.2	10.4–13.8
1	0.1	2.89 [2.41, 3.34]	–	1.508	10.0	8.5–11.2
1	0.3	1.80 [1.53, 2.04]	✓	1.102	11.6	9.9–13.1
1	1	0.88 [0.74, 1.04]	✓	1.023	12.1	10.3–13.6
1	3	0.55 [0.41, 0.68]	–	1.013	12.2	10.3–13.7
1	∞	0.37 [0.28, 0.48]	–	1.008	12.2	10.4–13.7

Evaluated across 20 (ρ, k) combinations under $S1$ heterogeneity ($v_R = 0.0150$). A cell is compatible with the empirical counting slope if its 95% Monte Carlo range overlaps the observed 95% confidence interval $[0.81, 1.73]$ (14 of 20 cells compatible). Across all compatible cells, a/v_R stays tightly within 0.996–1.102, yielding an implied transmission relative SD of 11.6–12.3% (9.9–13.8% propagated). Sensitivity across 180 grid decisions: 8 change under 90% limits, 20 change under 99% limits, and 8 change under ± 2 MCSE tolerance; exactly 0 cells pass all six tests under all rules.

comes from the estimator and not from the interval, because the mean of a under S8 is 0.857 times the total. Under S9, which adds crossed state and national-week dependence, the intervals cover the estimator means in 0.998 (a), 1.000 (b) and 0.998 (b/a) of panels and the true total in 0.963 (0.009), with the mean of a at 0.889 times the total. The bootstrap intervals are therefore wide rather than calibrated.

The two-way sandwich meets the decision rule, and it is the calibrated interval reported for a, b and b/a (Table 11). Its coverage of the estimator mean for a is 0.993 (0.004) under S1, 0.925 (0.013) under S6 and S8, and 0.983 (0.007) under S9; for b it is 0.917–0.945 and for b/a 0.915–0.970. The rule was applied to a; because the coverage for b and b/a falls slightly below 0.925 under S6, the bootstrap intervals are reported alongside the sandwich intervals for all three summaries. Under S6 and S8 the coverage of a equals the threshold, so the rule is met without margin. The mean sandwich standard error of a is 0.94–1.40 times the Monte Carlo standard deviation. Because the sandwich interval is centred on a biased estimator under S8, it covers the true total in only 0.42 of S8 panels, against 0.93 under S6. The scaled bootstrap covers the estimator mean for a in 0.915–0.990 of panels and would have failed the rule under S8.

The season test needs the same calibration. Its Wald statistic with the sandwich covariance is liberal at the χ^2_3 critical value, rejecting a true null in 0.195 (0.020) of S6 panels and 0.217 (0.021) of S8 panels, and the simulated 95th percentile of the null statistic is 6.8–16.5 across generators against 7.8 for χ^2_3. With the bootstrap covariance the statistic is very conservative, with a simulated null 95th percentile of 1.3–2.7 and no rejection at the χ^2_3 critical value. The observed season statistics are referred to these simulated null distributions in Section 5.3.

To resolve reliance on synthetic generators whose panels vary less than the data (observed sandwich standard error of a is 1.7–2.9 times generator Monte Carlo standard deviations), Table 11 additionally provides non-parametric calibrations computed directly on the empirical data. A Wild cluster bootstrap with Webb 6-point weights at the state cluster level yields a calibrated null 95th percentile of 9.8 and $p = 0.013$ ($SE = 0.002$). A within-state season permutation test, which preserves all intra-state dependent correlation while permuting season labels, gives a null 95th percentile of 17.1 and $p = 0.045$ ($SE = 0.005$). Both non-parametric references confirm rejection of equal season intercepts at the 5% level without relying on generator smoothness assumptions.

Table 7 Known-truth audit of the intercept on the real R_A design against transmission and total variance by generator

Generator	D median [P10, P90]	Transmission v_R	Total v_R + v_H	Intercept a (MC SE)	a / v_R	a / (v_R + v_H)	True-error intercept	Week-centred a reduction
Observed (R_A, GI 3.6)	45 [10, 389]	unknown	unknown	0.0150 [0.0101, 0.0206]	–	–	–	0.35
S0 Poisson, no heterogeneity	45 [10, 281]	0	0	0.0000 (0.0000)	$v_R = 0$	total 0	0.00003 [-0.00010, 0.00019]	not defined
S1 iid heterogeneity, Poisson counts	45 [11, 263]	0.0150	0.0150	0.0151 (0.0000)	1.01 [0.93, 1.09]	1.01 [0.93, 1.09]	0.00003 [-0.00010, 0.00021]	-0.01 [-0.04, 0.01]

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Generator	D median [P10, P90]	Transmission v_R	Total v_R + v_H	Intercept a (MC SE)	a / v_R	a / (v_R + v_H)	True-error intercept	Week-centred a reduction
S2 NB offspring + binomial reporting, no heterogeneity	45 [10, 324]	0	0	0.0001 (0.0000)	v_R = 0	total 0	0.00002 [-0.00013, 0.00018]	not defined
S3 iid heterogeneity + NB/reporting	45 [10, 298]	0.0147	0.0147	0.0150 (0.0001)	1.02 [0.91, 1.13]	1.02 [0.91, 1.13]	0.00003 [-0.00011, 0.00020]	-0.02 [-0.05, 0.01]
S4 state + week heterogeneity, Poisson counts	45 [11, 315]	0.0135	0.0135	0.0150 (0.0000)	1.12 [1.04, 1.19]	1.12 [1.04, 1.19]	0.00002 [-0.00006, 0.00015]	0.62 [0.58, 0.66]
S5 state + week heterogeneity + NB/reporting	45 [10, 324]	0.0130	0.0130	0.0149 (0.0001)	1.15 [1.06, 1.27]	1.15 [1.06, 1.27]	0.00003 [-0.00007, 0.00017]	0.59 [0.53, 0.64]
S2P Poisson offspring + binomial reporting, no heterogeneity	45 [10, 293]	0	0	0.0000 (0.0000)	v_R = 0	total 0	0.00002 [-0.00008, 0.00015]	not defined
S6 S3 + state-week reporting shock (half of a)	45 [10, 307]	0.0074	0.0147	0.0148 (0.0001)	2.02 [1.71, 2.29]	1.01 [0.87, 1.14]	0.00746 [0.00590, 0.00877]	-0.01 [-0.05, 0.02]
S7a S3 with scale-dependent heterogeneity, γ 0.25	45 [10, 296]	0.0217	0.0217	0.0148 (0.0001)	0.68 [0.60, 0.78]	0.68 [0.60, 0.78]	0.00003 [-0.00011, 0.00020]	-0.03 [-0.08, 0.02]
S7b S3 with scale-dependent heterogeneity, γ 0.5	44 [11, 286]	0.0346	0.0346	0.0150 (0.0001)	0.43 [0.37, 0.50]	0.43 [0.37, 0.50]	0.00002 [-0.00013, 0.00021]	0.00 [-0.08, 0.06]
S8 Combo S6 shock + S7a scale-dep (γ 0.25)	45 [10, 313]	0.0084	0.0173	0.0148 (0.0001)	1.75 [1.47, 2.04]	0.85 [0.72, 0.99]	0.00896 [0.00729, 0.01093]	-0.02 [-0.06, 0.02]

200 synthetic panels per generator on the real 1748-row R_A design (growth factor g_0 = 1.407); analyst scale D = simulated C_t and the filter $C_t \geq 5$ applied to each panel; D quantiles are Monte Carlo means over panels. The calibration targets, which are not tests, are the median retained D (count scale 0.797-0.844, all generators) and, for the generators with heterogeneity (S1, S3, S4, S5, S6, S7a, S7b, S8), the Monte Carlo mean of a, which equals the observed a by construction. All other columns are tests. The upper tail of D is not matched (generator P90 263-324, observed 389). Post-stratifying retained rows to the observed deciles of D changes a / (v_R + v_H) from 1.007 to 1.008 (S1), 1.115 to 1.079 (S4), 1.009 to 1.004 (S6) and 0.684 to 0.665 (S7a). v_R is the relative variance of the target R_i on retained rows, and v_H the current-value error caused by reporting shocks (S6, S8 only). Ratios are ratios of panel means with the 2.5-97.5% range of panel ratios; MC SE = SD/sqrt(panels). The true-error intercept is the affine fit of ((x - R_i)/$\bar{x}$)^2. Week-centred reduction = 1 - a_{week}/a, not defined without transmission heterogeneity. The observed a uses the primary bootstrap (2,000 draws). Shock log-SD s = 0.122 (S6) and 0.133 (S8). Under S6 the true-error intercept / v_H is 1.02 [0.83, 1.19]; lag-1 correlation of the shock term -0.50 [-0.53, -0.46]. The transmission relative SD implied by the observed a under each generator's a / v_R ranges from 9% to 19%.

Table 8 Known-truth audit of b/a, slope references, thinning and the in-sample contrast by generator

Generator	b/a	b/b_P	b_w/b_P (state FE)	Pooled / within slope	Thinning c/b_P	In-sample contrast	Quantile of observed contrast (MC SE)
Observed	37.4 [18.4, 72.4]	1.257 [0.812, 1.734]	0.852 [0.237, 1.475]	1.48 [0.99, 3.66]	0.909 [0.810, 0.999]	-0.0014	–
S0 Poisson, no heterogeneity	—	0.930 [0.853, 1.005]	0.896 [0.794, 1.021]	1.04 [0.95, 1.13]	0.970 [0.946, 1.000]	-0.0004 [-0.0028, 0.0018]	0.195 (0.013)
S1 iid heterogeneity, Poisson counts	25.9 [19.4, 32.9]	0.855 [0.686, 1.020]	0.819 [0.617, 1.026]	1.05 [0.94, 1.21]	0.950 [0.912, 0.986]	0.0119 [0.0035, 0.0213]	0.001 (0.001)
S2 NB offspring + binomial reporting, no heterogeneity	—	1.225 [1.128, 1.321]	1.206 [1.064, 1.342]	1.02 [0.95, 1.10]	0.929 [0.898, 0.970]	-0.0003 [-0.0033, 0.0031]	0.152 (0.011)
S3 iid heterogeneity + NB/reporting	36.3 [27.4, 44.4]	1.143 [0.983, 1.308]	1.123 [0.899, 1.333]	1.02 [0.93, 1.14]	0.879 [0.828, 0.927]	0.0164 [0.0044, 0.0281]	0.001 (0.001)
S4 state + week heterogeneity, Poisson counts	22.2 [17.0, 27.7]	0.734 [0.602, 0.871]	0.649 [0.486, 0.819]	1.14 [0.99, 1.37]	0.997 [0.964, 1.035]	0.0123 [0.0032, 0.0229]	0.000 (0.000)
S5 state + week heterogeneity + NB/reporting	32.6 [24.7, 40.0]	1.025 [0.866, 1.201]	0.953 [0.743, 1.142]	1.08 [0.94, 1.25]	0.925 [0.875, 0.981]	0.0239 [0.0128, 0.0363]	0.000 (0.000)
S2P Poisson offspring + binomial reporting, no heterogeneity	—	0.797 [0.732, 0.863]	0.774 [0.684, 0.865]	1.03 [0.96, 1.13]	0.974 [0.953, 1.004]	-0.0004 [-0.0037, 0.0022]	0.171 (0.012)
S6 S3 + state-week reporting shock (half of a)	36.4 [28.2, 47.6]	1.144 [0.970, 1.342]	1.141 [0.935, 1.354]	1.01 [0.90, 1.12]	0.869 [0.814, 0.919]	0.0176 [0.0048, 0.0337]	0.003 (0.002)
S7a S3 with scale-dependent heterogeneity, γ 0.25	51.4 [39.5, 65.9]	1.564 [1.337, 1.787]	1.484 [1.249, 1.759]	1.06 [0.95, 1.18]	0.914 [0.861, 0.986]	-0.0003 [-0.0112, 0.0106]	0.495 (0.016)
S7b S3 with scale-dependent heterogeneity, γ 0.5	77.5 [58.4, 99.7]	2.251 [1.975, 2.582]	2.058 [1.656, 2.493]	1.10 [0.97, 1.21]	0.922 [0.862, 0.980]	-0.0142 [-0.0225, -0.0046]	0.987 (0.004)
S8 Combo S6 shock + S7a scale-dep (γ 0.25)	42.1 [30.8, 58.2]	1.306 [1.081, 1.520]	1.278 [1.043, 1.512]	1.02 [0.92, 1.15]	0.881 [0.810, 0.947]	0.0108 [-0.0021, 0.0290]	0.069 (0.008)

Table 9 Declared known-truth tests by generator under one decision rule

Generator	b/b_P	b/a	Thinning c/b_P	Pooled / within slope	Week-centring reduction	Score contrast	Passed
Observed	1.26	37.4	0.91	1.48	34.7	-0.0014	
S0	fail (0.85 to 1.01)	n.d.	fail (0.95 to 1.00)	fail (0.95 to 1.13)	n.d.	pass (0.195)	1/4
S1	fail (0.69 to 1.02)	fail (19.38 to 32.86)	fail (0.91 to 0.99)	fail (0.94 to 1.21)	fail (-4.0 to 0.8)	fail (0.001)	0/6
S2	pass (1.13 to 1.32)	n.d.	pass (0.90 to 0.97)	fail (0.95 to 1.10)	n.d.	pass (0.152)	3/4
S2P	fail (0.73 to 0.86)	n.d.	fail (0.95 to 1.00)	fail (0.96 to 1.13)	n.d.	pass (0.171)	1/4
S3	pass (0.98 to 1.31)	pass (27.41 to 44.44)	pass (0.83 to 0.93)	fail (0.93 to 1.14)	fail (-5.3 to 0.8)	fail (0.001)	3/6
S4	fail (0.60 to 0.87)	fail (17.04 to 27.65)	fail (0.96 to 1.04)	fail (0.99 to 1.37)	fail (58.0 to 65.7)	fail (0.000)	0/6
S5	fail (0.87 to 1.20)	pass (24.66 to 40.02)	pass (0.87 to 0.98)	fail (0.94 to 1.25)	fail (53.5 to 64.2)	fail (0.000)	2/6
S6	pass (0.97 to 1.34)	pass (28.17 to 47.60)	pass (0.81 to 0.92)	fail (0.90 to 1.12)	fail (-5.0 to 1.6)	fail (0.003)	3/6
S7a	fail (1.34 to 1.79)	fail (39.46 to 65.92)	pass (0.86 to 0.99)	fail (0.95 to 1.18)	fail (-7.5 to 1.9)	pass (0.495)	2/6
S7b	fail (1.98 to 2.58)	fail (58.42 to 99.67)	pass (0.86 to 0.98)	fail (0.97 to 1.21)	fail (-7.5 to 5.6)	fail (0.987)	1/6
S8	pass (1.08 to 1.52)	pass (30.81 to 58.24)	pass (0.81 to 0.95)	fail (0.92 to 1.15)	fail (-6.1 to 2.3)	pass (0.069)	4/6
S9	pass (1.04 to 1.48)	pass (29.45 to 51.74)	pass (0.82 to 0.94)	fail (0.92 to 1.17)	fail (12.5 to 25.9)	pass (0.057)	4/6

Pass or fail under the decision rule of Section 4. Brackets give the Monte Carlo 2.5–97.5% range or, for the power-minus-affine score contrast, the quantile of the observed value pooled over five streams. The week-centring reduction is in per cent. n.d., not defined because the intercept is essentially zero. The pooled/within slope ratio is b/b_w, with b_w the state fixed-effects slope. S9 is S8 with half of the transmission heterogeneity carried by a national calendar-week component.

b_P is the independent-Poisson reference computed on each panel. S2, S3, S5, S6, S7 and S8 use negative-binomial offspring (k = 0.3) with binomial reporting probability 0.25; S2P uses Poisson offspring with the same reporting. Every column is a test (none is a calibration target). b/a undefined when a ≈ 0 (S0, S2 and S2P, whose true relative variance is 0), shown as —. Pooled / within slope = (b/b_P)/(b_w/b_P) with state fixed effects. The observed row uses the primary bootstrap (2,000 draws); its pooled / within slope is the slope ratio of the fixed-effects decomposition. Thinning uses 100 panels × 20 shared-count binomial draws per q and the least-squares rate through the origin. Contrast = in-sample power minus affine mean score (lower is better, so positive values favour the affine form). The last column gives the share of synthetic panels with a contrast at or below the observed value, pooled over 5 independent random streams of 200 panels each, with its binomial MC SE; the SD of the per-stream quantiles is 0.000-0.039.

Table 10 Coverage of the state-week bootstrap intervals on known-truth panels

Generator	a covers $v_R + v_H$	a covers E(a)	b covers E(b)	b/a covers E(b)/E(a)	E(a) / ($v_R + v_H$)	Bootstrap SD / MC SD (a, b, b/a)
S1 iid heterogeneity, Poisson counts	1.000 (0.000)	1.000 (0.000)	0.995 (0.004)	1.000 (0.000)	1.004	2.52, 1.69, 2.01
S6 S3 + state-week reporting shock (half of a)	0.998 (0.002)	0.998 (0.002)	0.995 (0.004)	0.998 (0.002)	1.014	1.76, 1.61, 1.72
S8 Combo S6 shock + S7a scale-dep (γ 0.25)	0.892 (0.015)	1.000 (0.000)	0.998 (0.002)	1.000 (0.000)	0.857	1.71, 1.76, 1.82
S9 S8 with half of transmission heterogeneity national-week	0.963 (0.009)	0.998 (0.002)	1.000 (0.000)	0.998 (0.002)	0.889	1.92, 1.78, 1.94

400 known-truth panels per generator, each analysed as in the application with a gamma FGLS affine fit on retained rows and 200 state × calendar-week pigeonhole bootstrap draws (L = 1), percentile 95% intervals. Nominal coverage 0.95; binomial MC SE in parentheses (0.000 when every panel is covered; the one-sided 95% lower bound is then 1 - 3/400 = 0.993). E(.) is the Monte Carlo mean of the estimator over panels, so the 'covers E' columns isolate the calibration of the interval from the bias of the estimator; 'covers $v_R + v_H$' also includes that bias. The last column is the mean bootstrap SD over the Monte Carlo SD of the point estimates. In every generator all bootstrap draws converged and none had a ≤ 0.

Table 11 Coverage of the two-way cluster-robust interval and calibration of the season test by generator and non-parametric reference

	Coverage of the estimator mean (nominal 0.95)					Season test, 3 df			
Generator	a	b	b/a	SE ratio for a	Scaled bootstrap, a	Size at $\chi^2_{3,0.95}$	Null 95th percentile	Calibrated p, W_{CGM}	Calibrated p, W_{boot}
S1	0.993 (0.004)	0.945 (0.011)	0.970 (0.009)	1.40	0.990 (0.005)	0.043 (0.010)	6.8	0.013 (0.006)	< 0.005
S6	0.925 (0.013)	0.917 (0.014)	0.915 (0.014)	1.00	0.930 (0.013)	0.195 (0.020)	16.5	0.045 (0.010)	< 0.005
S8	0.925 (0.013)	0.938 (0.012)	0.922 (0.013)	0.94	0.915 (0.014)	0.217 (0.021)	14.5	0.028 (0.008)	< 0.005
S9	0.983 (0.007)	0.927 (0.013)	0.963 (0.009)	1.22	0.965 (0.009)	0.072 (0.013)	8.8	0.005 (0.004)	< 0.005
Non-parametric calibrations on real data:							9.8	0.013 (0.002)	–
Wild cluster bootstrap (Webb 6-point, state-level clusters, 2,000 draws)							17.1	0.045 (0.005)	–
Within-state season permutation test (season exchangeability, 2,000 draws)							7.8	0.0005	0.138
Observed	a 0.0150 (0.0109–0.0191); b/a 37.4 (24.1–58.3)					$W_{CGM} = 17.8$; $W_{boot} = 5.52$			

Coverage over 400 panels per generator, with binomial Monte Carlo standard errors in brackets. The two-way interval is the state-by-week cluster-robust sandwich defined in Section 4. The SE ratio divides the mean sandwich standard error by the Monte Carlo standard deviation of a. The scaled bootstrap shrinks the pigeonhole interval for a by the smallest bootstrap-to-Monte-Carlo ratio among the other three generators. The season test is the Wald test of equal log season intercepts with a common slope ($W_{CGM} = 17.8$). Its size is the rejection rate at the χ^2_3 critical value with each season centred at its Monte Carlo mean. Non-parametric calibrations (Wild cluster bootstrap with Webb 6-point weights and within-state season permutation) operate directly on the empirical data and account for the observed dependent variation.

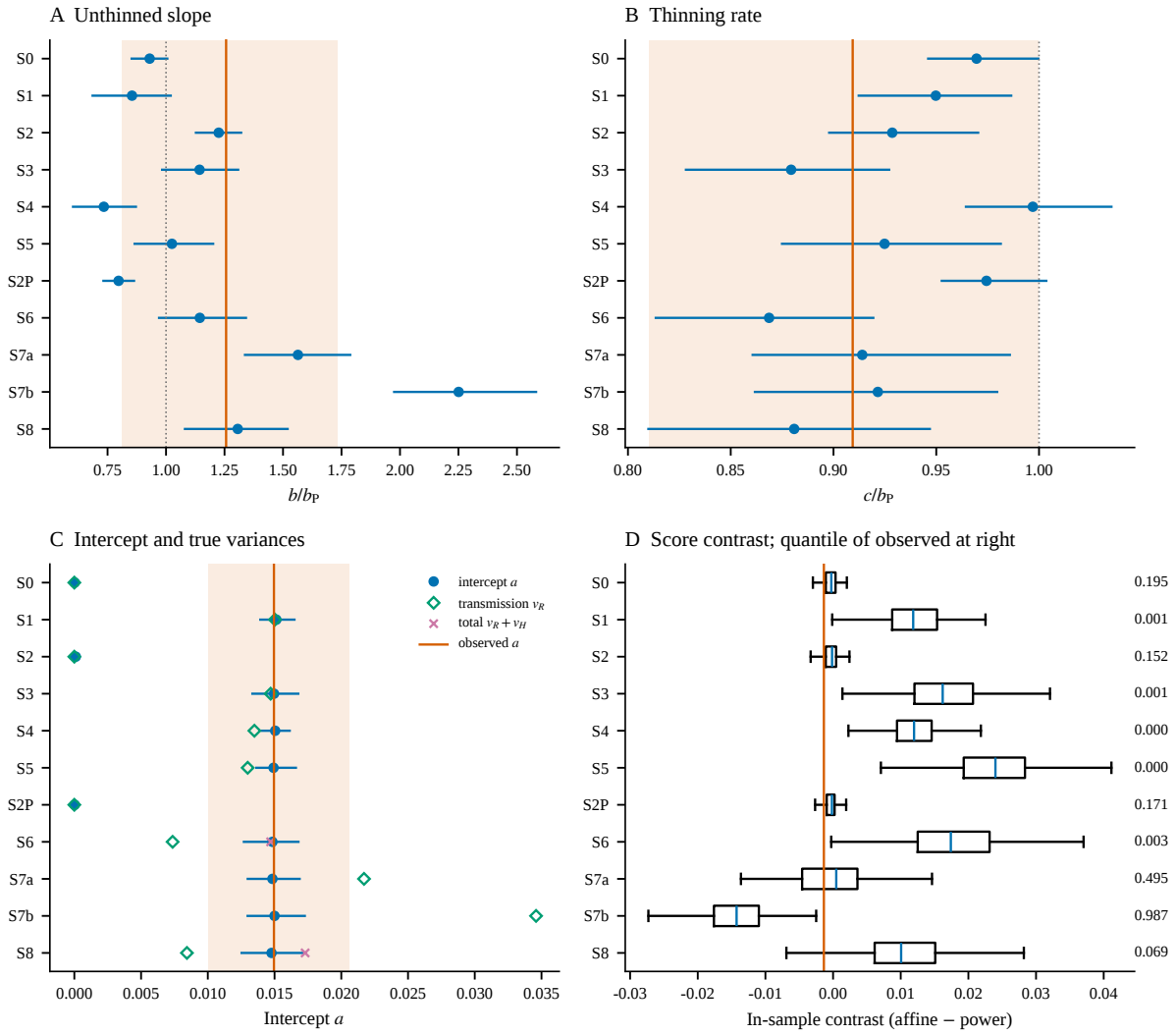


Figure 2 Known-truth audit of the dispersion diagnostics on the real R_A design, with simulated counts aligned to the observed median. Panel A shows the counting slope relative to the independent-Poisson reference, b/b_P, by generator. Panel B shows the thinning rate relative to the same reference, c/b_P. Panel C shows the fitted intercept a against the true transmission relative variance and the true total, which differ only under S6 and S8. Panel D shows the in-sample score contrast between the affine and power-law fits, with the quantile of the observed value among synthetic panels at the right; boxes span the interquartile range and whiskers 1.5 times it. In panels A–C points are means over 200 panels per generator (100 panels for thinning) and bars are 2.5–97.5% Monte Carlo ranges. The vertical line and shaded band mark the observed value and its bootstrap 95% confidence interval; dotted lines mark 1. S0, Poisson; S1, independent heterogeneity; S2, negative binomial offspring with reporting; S2P, Poisson offspring with reporting; S3, S1 with S2; S4, state and national-week heterogeneity; S5, S4 with S2; S6, S3 with week-to-week reporting shocks; S7a and S7b, S3 with heterogeneity that shrinks with count; S8, S3 with S7a heterogeneity and S6-type reporting shocks. Generator S9 is reported in Tables 9 and 10.

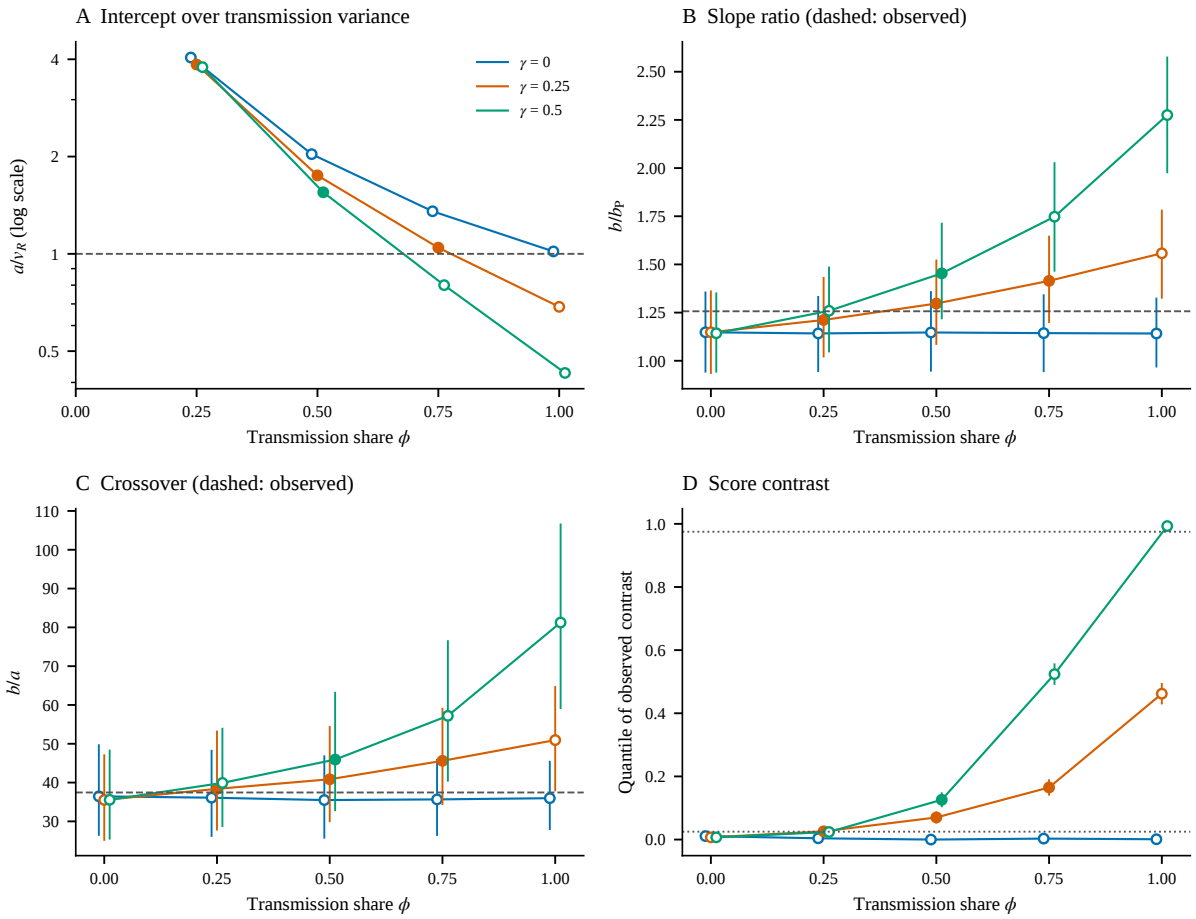


Figure 3 Known-truth grid over the transmission share ϕ of the total relative variance and the scale dependence γ of heterogeneity, in calibrated cells with 1,000 panels each. (A) Intercept relative to the true transmission relative variance, on a log scale. (B) Slope ratio b/b_P. (C) Crossover count b/a. (D) Quantile of the observed score contrast among synthetic panels. Dashed lines mark the observed values; in (A) the dotted line marks 1. Bars are 2.5–97.5% Monte Carlo ranges.

5 Application to US influenza hospital admissions

5.1 Count-independent relative dispersion

For R_A the pooled affine fit gave $a = 0.0150$ (calibrated two-way 95% confidence interval 0.0109–0.0191) and $b = 0.560$ (0.430–0.690); the conservative bootstrap intervals are 0.0101–0.0206 and 0.362–0.769 (primary row of Table 12; Table 11). The count-independent relative dispersion a corresponds to a relative SD of about 12% around the pooled mean of R_A. It is not by itself an estimate of transmission variance, because the audit shows that the intercept recovers different quantities under different mechanisms (Section 4.4). Read through the recovery ratio of each audited generator with heterogeneity, the observed intercept implies a transmission relative SD from 9% (S6) to 19% (S7b). Under reporting shocks of the S8 and S9 kinds the intercept overstates transmission variance by a factor of 1.75–1.84 and implies 9%, whereas under scale-dependent heterogeneity (S7a, S7b) it understates it. In the share grid no cell passes all six tests. Restricted to the cells that pass the slope ratio, the crossover and the contrast, the implied transmission relative SD is 6–14% at the observed a and 5–16% when the uncertainty in a is propagated, and the share of a carried by transmission is not identified (Tables S1 and S2). These figures are conditional on the audited mechanisms, none of which passes every test (Table 9). The renewal growth statistic G is reported in Appendix Table S3 as an un-audited descriptive comparison on the weighted denominator D_t.

Week centring, which replaces the pooled mean by the national mean of each calendar week, reduced a by 34.7% (6.0–51.9) for R_A and by 33.3% (6.8–51.6) for G (Table S8; Fig. S2). Leaving the own state out of the weekly centre gave nearly the same values. Season means and season-specific linear trends left a almost unchanged. In the audit, week centring removed almost nothing when transmission heterogeneity was independent across states (-2.8% to 0.4% in S1, S3, S6, S7a and S7b) and removed 61.8% (S4) and 58.6% (S5) when half of it was national. The observed reduction lies between these values, which is consistent with a partly synchronous national component of intermediate size. Under S9, in which half of the transmission heterogeneity is national and reporting shocks carry half of the variance target, the reduction is 19.1% (12.5–25.9), below the observed value. Adjusting for hospital reporting participation changed a by 0.00029 (-0.00032 to 0.00085) (Table S9). Off-season summer weeks, outside the analysis windows, are described in

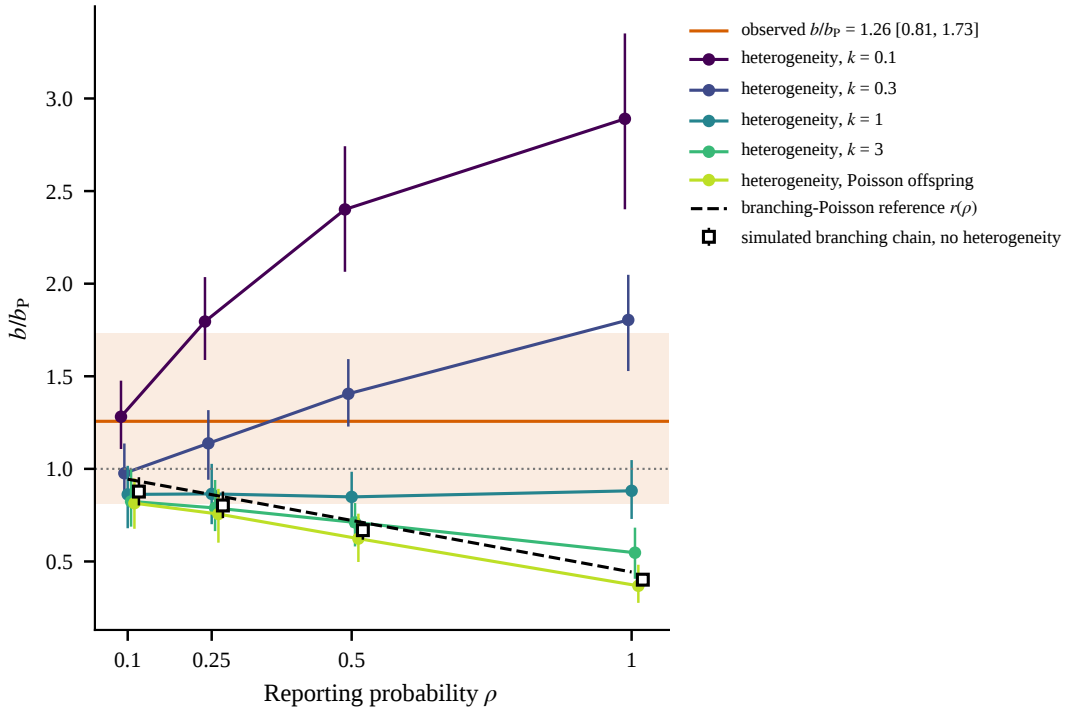


Figure 4 Expected counting slope relative to the independent-Poisson reference, b/b_P, against reporting probability ρ for five offspring laws. Every simulated panel includes independent transmission heterogeneity as in generator S1 and is count-aligned as in the audit; points are means over 100 synthetic panels on the real R_A design and bars are Monte Carlo 2.5–97.5% ranges. The Poisson-offspring curve declines with ρ because, given the realised parent count, the denominator carries only the binomial reporting noise $(1-\rho)/C_t$, as in the branching-Poisson reference. The dashed line is the analytic branching-Poisson reference $r(\rho)$ of Eq. (17) and open squares are simulated branching chains with Poisson offspring and no heterogeneity (Table S4). The horizontal line and shaded band mark the observed b/b_P and its 95% confidence interval.

Table S14. Reporting was voluntary only in summer 2024, when median participation was 45.6% against 92.3–94.2% in the other two summers. The interquartile range of $\log G$ was then 0.67 against 0.49, a descriptive contrast.

5.2 Counting slope

The pooled counting slope was 1.26 (0.81–1.73) times the independent-Poisson reference, 26% above it with an interval that includes 1. Against the branching-Poisson reference of Eq. (17) the ratio was 1.46 (0.94–2.02) at $\rho = 0.25$ and 2.84 (1.83–3.94) at $\rho = 1$ relative to the analytic reference; relative to the simulated reference, which shares the finite-sample bias of the fitting procedure, the ratios are about 1.57 and 3.13 (Table S4). The excess over a branching-Poisson reference therefore depends on the reporting probability, which the slope itself does not identify.

Within-unit estimates were lower and less precise (Table S10; Fig. 5B). A state fixed-effects fit gave 0.85 (0.24–1.47) b_P by iterated weighting and 1.02 (0.51–1.53) by a one-step weighted fit; a damped iterated two-way fit gave 0.93 (0.09–1.48) b_P, with 1,962 of 2,000 bootstrap draws converging, and its one-step counterpart 1.08 (0.47–1.66). The undamped two-way iteration oscillated between two values and did not converge. A week fixed-effects fit gave 1.42 (0.96–1.89) b_P.

The pooled and state fixed-effects fits can also be compared through their crossover counts. The cross-count ratio, pooled b/a divided by state fixed-effects b over mean state intercept, was 2.16 (0.99–7.64). It factorises exactly into a slope ratio, pooled b over state fixed-effects b, of 1.48 (0.99–3.66), and an intercept ratio, mean state intercept over pooled a, of 1.47 (0.99–2.29); the slope ratio accounts for half of the log cross-count ratio (0.50, 0.31–0.74) (Table S11). Under single-week resampling all three intervals include 1; under two-week blocks all three still do, and under three-week blocks the three lower limits are 1.01–1.03. In the audit, generator S4, in which all states share one counting process, gave a pooled-to-within slope ratio of 1.14 (Monte Carlo range 0.99–1.37) (Table 8). Structured heterogeneity alone thus raises the pooled slope above the within-state slope. The observed slope ratio lies above the S4 value, but its interval overlaps the S4 range, so between-state differences in counting remain unresolved.

The counting slope also does not identify reporting probability or offspring dispersion. With independent transmission heterogeneity of the S1 kind, the expected b/b_P ranged from 0.37 (Poisson offspring, $\rho = 1$) to 2.89 ($k = 0.1$, $\rho = 1$) across four reporting probabilities and five offspring laws (Fig. 4; Table S12). The observed interval contains the expected value in 10 of the 20 cells, namely all five offspring laws at $\rho = 0.1$, $k = 1$ at every reporting probability and $k = 0.3$ at $\rho = 0.25$ and 0.5. Very different combinations of under-reporting and superspreading are compatible with the same slope, and heterogeneity of the S4 kind would lower the expected slopes (S4 0.73 against S1 0.85 in Table 8).

5.3 Dispersion-to-counting crossover count

The dispersion-to-counting crossover count was $b/a = 37.4$ admissions per week for R_A, with a calibrated two-way interval of 24.1–58.3 and a bootstrap interval of 18.4–72.4; the sensitivity intervals below are bootstrap intervals. The point estimate was 49.7 (25.0–96.6) when rows required $C_t \geq 10$ (1,535 rows), 52.8 (19.0–122.9) at $C_t \geq 20$ (1,203 rows) and 53.1 (-8.4 to 180.1) at $C_t \geq 50$ (776 rows), where the ratio was nonpositive in 4.8% of draws. A continuity correction of 1.0 instead of 0.5 gave 33.1 (16.3–66.3) and a window threshold of 1.10 gave 38.6 (20.6–70.3) on 1,564 rows (Table 12; Fig. 5A). The value 37.4 is an in-sample descriptive crossover for this panel and window rule, not a sample-size rule for surveillance.

Table 12 Sensitivity of the pooled loss model to filters, continuity correction, window rule and season (bootstrap 95% confidence interval in brackets)

Specification	n	a	b	b/a	b/b_P
Primary (C_t, $D_t \geq 5$; cc 0.5; growth rule 1.05)	1748	0.0150 [0.0101, 0.0206]	0.560 [0.362, 0.769]	37.4 [18.4, 72.4]	1.26 [0.81, 1.73]
Counts ≥ 10	1535	0.0136 [0.0090, 0.0188]	0.677 [0.436, 0.938]	49.7 [25.0, 96.6]	1.54 [0.99, 2.14]
Counts ≥ 20	1203	0.0133 [0.0081, 0.0189]	0.701 [0.332, 1.092]	52.8 [19.0, 122.9]	1.63 [0.76, 2.52]
Counts ≥ 50	776	0.0129 [0.0073, 0.0191]	0.686 [-0.155, 1.479]	53.1 [-8.4, 180.1] (b/a ≤ 0 in 4.8% of draws)	1.64 [-0.38, 3.59]
Continuity correction 1.0	1748	0.0154 [0.0102, 0.0214]	0.510 [0.326, 0.728]	33.1 [16.3, 66.3]	1.15 [0.74, 1.64]
Growth-window rule 1.02	1748	0.0150 [0.0100, 0.0207]	0.560 [0.366, 0.779]	37.4 [18.8, 70.0]	1.26 [0.82, 1.74]
Growth-window rule 1.10	1564	0.0141 [0.0097, 0.0187]	0.543 [0.357, 0.760]	38.6 [20.6, 70.3]	1.26 [0.83, 1.75]
Season 2022-23	365	0.0137 [0.0067, 0.0210]	0.697 [0.350, 1.083]	50.9 [19.8, 134.8] (5 draws not converged)	1.62 [0.80, 2.50]
Season 2023-24	499	0.0062 [0.0025, 0.0141]	0.702 [0.403, 0.963]	113.7 [30.2, 315.1] (2 draws not converged)	1.51 [0.85, 2.10]
Season 2024-25	356	0.0127 [0.0058, 0.0232]	0.583 [0.161, 1.049]	45.9 [8.2, 147.2]	1.38 [0.36, 2.50]
Season 2025-26	528	0.0225 [0.0085, 0.0355]	0.353 [0.129, 0.718]	15.7 [4.1, 71.6]	0.78 [0.28, 1.61]

Each row refits the affine FGLS on the subset or variant with a state-week bootstrap; the primary row is the primary bootstrap (2,000 draws, identical to Table S3), the other rows use 2,000 draws each. Where b/a ≤ 0 (b ≤ 0) in more than 2.5% of draws its percentile interval is unstable; draws whose FGLS fit did not converge are dropped and counted. The growth-window rule 1.02 selects the same 1,748 rows as the primary rule, so its row differs from the primary row only through the bootstrap draws.

Fitted separately, the four seasons gave 50.9, 113.7, 45.9 and 15.7 on 365, 499, 356 and 528 rows. These values are descriptive, and each has a wide interval, from 4.1–71.6 in 2025–26 to 30.2–315.1 in 2023–24. A model with season-specific intercepts and a common slope ($b = 0.607$, 0.415–0.794) gave season crossover counts of 38.3, 85.7, 39.5 and 33.7 (Table S13). With a common slope, equal season intercepts are equivalent to equal season crossover counts. The test of equality is a Wald test on log intercepts, chosen because the intercepts are strictly positive and multiplicative differences are the quantity of interest. Referred to χ^2_3, the statistic with the inter-season bootstrap covariance gave $W = 5.52$ ($p = 0.138$). Cochran's Q, evaluated using diagonal inverse-variance weights without off-diagonal covariance, was 4.57 ($p = 0.206$, $I^2 = 34\%$). On the raw intercept scale the Wald test gave $p = 0.064$, and a test on separate season fits gave $p = 0.220$. These χ^2 references are not calibrated, because the bootstrap covariance is conservative (Section 4.4). Referred to the simulated null distributions of the four coverage generators, $W = 5.52$ exceeds every simulated null statistic (calibrated $p < 0.005$), and the statistic with the two-way sandwich covariance, $W_{\mathrm{CGM}} = 17.8$, has calibrated p values of 0.005–0.045 (Table 11). Non-parametric calibrations on the empirical data, which naturally incorporate the observed dependent variation, confirm the rejection: the Wild cluster bootstrap with Webb weights gives $p = 0.013$ and the within-state season permutation gives $p = 0.045$. Under all calibrated specifications the test rejects equal season intercepts at the 5% level. The difference comes mainly from the second season (2023–24), whose fitted intercept was 0.0071 against 0.0154–0.0180 in the others, which pushes its crossover count to 85.7. Under the uncalibrated χ^2_3 reference with the bootstrap covariance the test has limited power: reaching 80% power at the 5% level along the observed pattern requires the deviations to be 1.41 times larger, corresponding to a pattern-specific minimum detectable ratio of 3.71 between the highest and lowest season intercepts, against the observed 2.54. Even when differences are detected, the between-season variance itself can be estimated only to a relative SE of at least 0.82 with four seasons (Appendix B.3).

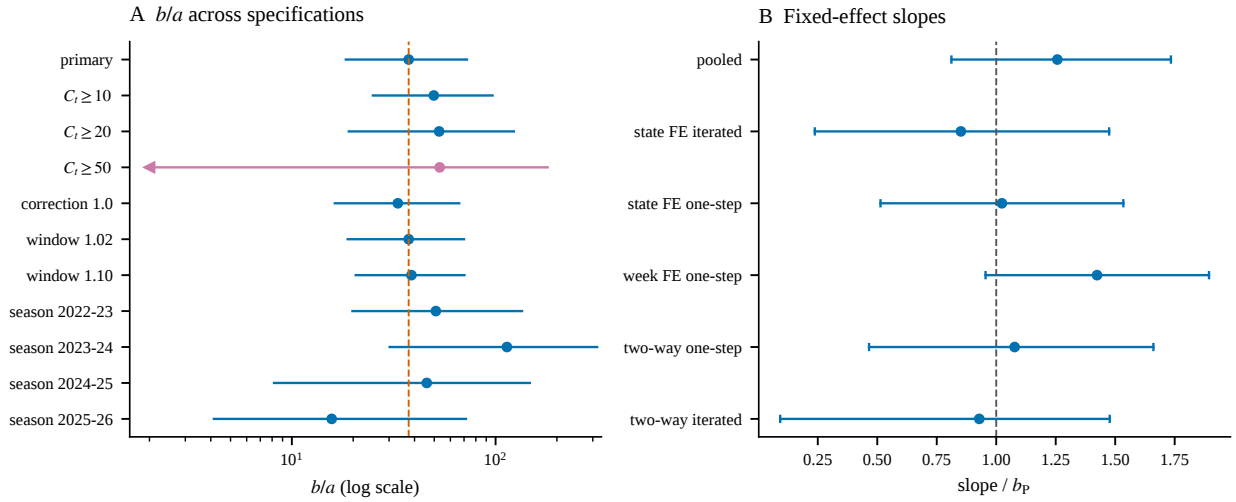


Figure 5 Sensitivity of the dispersion summaries for R_A. (A) Dispersion-to-counting crossover count b/a (log scale) under the primary specification, minimum-count filters, a continuity correction of 1.0, window thresholds of 1.02 and 1.10 and separate season fits; the dashed line marks the primary estimate. For $C_t \geq 50$ the ratio was nonpositive in more than 2.5% of draws, so the lower bound lies below the axis. (B) Counting slopes relative to the independent-Poisson reference for pooled, state, week and two-way fixed-effects fits; the dashed line marks 1. Bars are bootstrap 95% confidence intervals.

The interval choice has almost no effect on the ratio. Replacing the primary interval with a mean of 3.2 days changed b/a by 0.17 (-0.65 to 0.89), and the intrinsic generation-time estimate with standard deviation 2.1 days changed it by 0.49 (-2.55 to 3.04), with b/b_P 1.27 and 1.23 (Table S15). The interval rescales a and b together, so it affects the levels of the two summaries but not their ratio.

5.4 Shape of the dispersion curve

Held-out comparison of the affine model with a power law $\lambda C^{-\gamma}$ did not separate the two forms. Leaving out one state at a time, the power-minus-affine score difference was -0.0020 (paired standard error 0.0068), with 26 of 51 folds favouring the affine model. Leaving out one season at a time it was 0.0046 (0.0087), with 3 of 4 folds favouring the affine model (Tables S17 and S19; Fig. S3). The held-out state contrast nevertheless discriminates between mechanisms. On synthetic panels analysed with the same folds, generators with heterogeneity that does not depend on count favour the affine form by 0.011–0.024 on average, and none of their panels reaches a contrast as low as the observed one. The observed value is typical of the generators without heterogeneity and of S7a, and lies at the 0.050 and 0.055 quantiles of S8 and S9 (Table S18). Out of sample, as in sample, the data therefore weigh against count-independent heterogeneity without favouring either form as a law. The affine form is used here as a working summary, not as an established law for surveillance dispersion.

6 Discussion

The theoretical part of the paper makes one distinction precise. With fixed, known reporting, exogenous exposure and conditionally independent counting errors, the risk for the current value is A/m and the risk for the realised mean of the observed periods is $A/(Tm)$. The process mean adds the variance of the realised mean, which more cases within the same periods do not reduce. The count at which counting error equals between-period variance is the half-weight point of the corresponding shrinkage estimator, as in small-area estimation and disease mapping [14, 15]. It belongs with its estimator, target and sampling experiment and is not a property of a pathogen. The efficiency statement holds within a class of conditionally unbiased estimators with a known denominator. Information from other features of the epidemic curve [6, 46], independent reporting measurements or prior structure changes what can be learned, and estimators that accept bias in exchange for lower variance lie outside that class. Appendix B gives the corresponding forecast weight, the allocation of a fixed budget between periods and cases, and a lower bound on the precision with which a few seasons can estimate between-season variation.

The empirical part shows what can and cannot be read from a cross-sectional dispersion curve. Its intercept is a count-independent relative dispersion, the part of the variation in the statistic that does not scale with the count. With reporting fixed across weeks it is the relative variance of transmission across units and weeks; with week-to-week reporting shocks it also contains their contribution, and single-period moments cannot split the two (Proposition 2). For influenza admissions it is about 0.015. Under pure transmission heterogeneity, scanning the entire (ρ, k) parameter manifold compatible with the counting slope gives a tightly bound transmission relative standard deviation between 11.6% and 12.3% (9.9% and 13.8%

propagated); expanding across the full grid of reporting shocks and scale dependence yields a global envelope between 5.0% and 20.4%. Under fixed reporting this is the variation that a current-value analysis wants to resolve and that a process-mean analysis treats as sampling variation; under volatile reporting it is an upper bound on the error floor of the current value. The counting slope, its ratio to the independent-Poisson or branching-Poisson reference and the thinning response cannot distinguish overdispersed offspring, under-reporting and heterogeneous transmission. The audit produces slopes near the observed one under several combinations, and the grid over ρ and k admits values from heavy under-reporting with strong superspreading to near-complete reporting with moderate dispersion. The dispersion-to-counting crossover count was about 37 admissions per week for the pooled panel. Equal season intercepts were rejected under both simulated null distributions and non-parametric data calibrations (Wild cluster bootstrap $p = 0.013$; permutation $p = 0.045$), mainly because of a lower intercept in 2023–24, confirming seasonal transmission divergence across surveillance cycles. The crossover marks where counting noise stops dominating the cross-sectional dispersion of this statistic, not where additional cases stop being useful for a particular target.

6.1 Real-time estimation

All analyses used a retrospective snapshot in which counts are close to final. In real time, reporting delays leave the most recent weeks incomplete, and right-censoring biases the latest counts downwards in proportion to the delay distribution [47, 48]. The current-value target is affected first and most, because its risk is dominated by the last one or two periods, exactly those that are incomplete, whereas a realised or process mean over many periods dilutes them. The R_A statistic also needs C_{t+1}, so it is available only one week after its reference week, and estimates near the end of a series combine smaller effective counts with nowcasting uncertainty. Real-time renewal methods address this by modelling delays and the infection process jointly [2, 49, 50]. In the present framework, nowcasting adds a correction-error term to the counting term, and its variance grows as the delay distribution places more mass beyond the current week. A crossover count estimated from retrospective data will therefore overstate the precision available in real time.

Estimation windows raise a related issue. Widening a window in renewal estimators accumulates incidence while imposing constancy of the reproduction number over a longer interval [2, 3]. This trades temporal resolution for precision and is not equivalent to observing more independent periods; the averages in Table 2 instead give each period its own transmission value. Applying their formulas to overlapping windows requires explicit treatment of the overlap.

6.2 Reporting that depends on transmission

Proposition 2 assumes that the reporting probability is independent of transmission. In practice testing, care seeking and admission thresholds can change with epidemic activity [51, 52]. Write $z_t = p_t/\rho$ for the relative reporting probability. Given R_t, the estimator then has conditional bias

$$\mathrm{E}(\hat{R}_t \mid R_t) - R_t = [\mathrm{E}(z_t \mid R_t) - 1]\, R_t. \tag{19}$$

By Eq. (19), if reporting rises with transmission, so that $\mathrm{Cov}(z_t, R_t) > 0$, high-transmission periods are over-counted and low-transmission periods under-counted, the estimated swings exaggerate the true ones, and the fitted intercept overstates the relative dispersion of transmission. If reporting falls with transmission, for example when admission capacity saturates or testing is rationed at the peak, the estimates damp the swings and the intercept understates it. In both cases the process-mean estimate is biased by $\mathrm{Cov}(z_t, R_t)$, and the bias does not shrink with more cases. Neither direction can be detected from the counts alone; it requires external information on testing or admission practice. The participation adjustment used here addresses only facility participation, and it changed the intercept negligibly.

6.3 Choice of interval distribution

The transform in Eq. (15) needs an interval distribution. We kept the household serial interval of Cowling et al. [40] as the primary choice because it is directly estimated for influenza, widely used and fixed before the analyses reported here. The intrinsic generation-time estimate of Chan et al. [42], the more appropriate quantity for a renewal mapping, served as a sensitivity analysis. Serial intervals, intrinsic generation intervals and realised household generation intervals are different quantities [41], and the mapping from growth rate to reproduction number depends on which is used [39]. The levels of a and b change with the interval, but their ratio and the slope ratio to the independent-Poisson reference change by less than their sampling uncertainty, so the conclusions about what the diagnostics measure do not depend on this choice.

6.4 Identification and limitations

Separating the mechanisms behind the dispersion curve requires information that a single cross-section lacks. Temporal structure can separate persistent transmission variation from independent counting and reporting noise under explicit state-space or renewal models [30, 31, 53], and external measurements of reporting can constrain ρ and its variation. The shape of the curve narrows the candidates without identifying one. The observed power-minus-affine contrast lay at or below

the 0.3% quantile of every generator whose heterogeneity does not depend on count (S1 and S3–S6). Scale-dependent heterogeneity alone (S7a) reproduces the contrast but not the slope ratio. Scale-dependent heterogeneity combined with reporting shocks (S8) reproduces the intercept, the crossover count, the slope ratio and the contrast, and the share grid shows a family of such combinations. Within that family the intercept is between 1.04 and 3.85 times the transmission relative variance, so the share of a due to transmission remains unidentified. The held-out state contrast gives the same ordering: it is improbable under count-independent heterogeneity and lies in the lower tail of S8 and S9. No mechanism in the audit or the grid reproduces the gap between pooled and within-state slopes, and week centring is reproduced only where part of the heterogeneity is national, so the family is an approximation and not a description of the data-generating process. Reporting that varies with jurisdiction size would leave the same trace.

The growth windows are a fixed rule applied to four seasons of one surveillance system, and inference conditions on that selection; window thresholds of 1.02 and 1.10 left the crossover count at 37.4 and 38.6, and transfer of these values to other epidemic regimes would need data from those regimes. The analyses used a retrospective snapshot, so reporting delays and right-censoring of recent weeks were not modelled (Section 6.1). The audit generators are stylised, and their count alignment matched the median and lower tail of the observed counts more closely than the upper tail, although reweighting to the observed count deciles changed the intercept recovery by at most 3.2%. During major epidemic surges where counts enter the upper tail ($C > 389$), potential non-linear admission thresholds or capacity constraints could induce additional scale-dependent compression of apparent dispersion, underscoring the value of auditing across flexible tail geometries. The pigeonhole bootstrap intervals were conservative in the audit, with standard deviations 1.6–2.5 times the Monte Carlo values. The two-way sandwich intervals reported for a, b and b/a met the coverage rule, but only at its threshold under S6 and S8, and intervals for other quantities remain bootstrap intervals. While the parametric generators have less dependent variation than the data, non-parametric Wild cluster bootstrap and within-state permutation tests directly on the empirical data independently corroborate the seasonal divergence at the 5% level. Scanning the full (ρ, k) manifold confirms that the implied transmission relative standard deviation remains bounded within 9.9–13.8% across all slope-compatible cells under pure heterogeneity, and within 5.0–20.4% across mechanisms with reporting shocks. Intervals for iterated fits condition on convergence, and within-unit slopes were imprecise and sensitive to the weighting scheme. Facility participation measures only one aspect of ascertainment, and the comparison with a second reporting regime in Appendix B.5 is descriptive.

6.5 Checklist for surveillance practice

Four recommendations for reporting reproduction numbers follow from these results.

1. State the inferential target. An estimate may target the current value R_t, the realised mean $\overline{R}_T$ over the observed periods or the process mean $\overline{R}$. Precision statements of the form A/m apply to the first two under declared reporting assumptions and omit between-period variation for the third.

2. Inflate the counting error when the denominator is an observed count. The delta factor is $1 + (1-\rho)(\overline{R}^2 + v_R)/A$, which is 1.48 under the illustrative parameters of Section 2.3. At small counts the factor is larger, 2.42 at $m = 10$ in Table 3. The half-weight count of the shrinkage estimator moves from $m_\times$ to $m_\times + (1-\rho)(\overline{R}^2 + v_R)/v_R$, from 53.1 to 78.6 under the same parameters.

3. Qualify single-period estimates for reporting volatility. One period's counts cannot separate transmission changes from reporting shocks (Proposition 2), so rapid changes should be checked against testing volumes, reporting participation or temporal models before they are interpreted as transmission.

4. Treat dispersion fits as descriptive. The intercept a, the slope b and the crossover count b/a describe the observed statistic on the selected rows. They are not a transmission-variance estimate without an assumed mechanism, and b/a is not a sample-size threshold for surveillance.

7 Conclusion

The inferential target decides what additional cases buy. More cases shrink the counting error of the current value and of the realised mean of the observed periods, but they do not remove the between-period variation that limits estimation of the process mean. When reporting fluctuates from week to week, the fluctuation leaves an error floor on the current value, and single-period second moments cannot separate it from transmission variation. A known-truth simulation study on the real surveillance design shows that the intercept of a cross-sectional dispersion curve recovers transmission variance, the transmission-plus-reporting total or less than either, depending on the mechanism. Such diagnostics therefore describe a count-independent relative dispersion and the counting noise around it. Separating under-reporting, superspreading and structured transmission requires temporal models or external data on reporting.

Data and code availability

The analyses used the dataset Weekly Hospital Respiratory Data (HRD) Metrics by Jurisdiction, National Healthcare Safety Network, published by the U.S. Centers for Disease Control and Prevention under the identifier ua7e-t2fy [36], in the snapshot retrieved on 20 October 2026. The COVID-19 counts for mainland China are from the Johns Hopkins CSSE repository [54]. The data-processing code and analysis-window definitions are in the public repository https://github.com/wangchi2068/reproduction-number-targets-surveillance (commit `6bbb8458a5ba3a9ba644d755320df55dbe06746b`). Code, data extracts, simulation settings and result files for every reported analysis are distributed with this article; running `bash code/run_all.sh` from the distributed experiment directory regenerates every reported estimate, table and figure from the stored inputs. No individual-level or restricted data were used.

Declarations

CRediT authorship contribution statement

Contributions are reported under the CRediT taxonomy: conceptualization, methodology, software, formal analysis, investigation, data curation, validation, visualization, writing (original draft) and writing (review and editing).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate

Ethics approval and informed consent were not required as this study analyzed publicly available, de-identified aggregate surveillance data.

A Proofs

A.1 Count law and conditional information

The probability generating function of one negative binomial offspring count is $[1 + (R_t/k)(1 - z)]^{-k}$. Conditional independence raises this function to the power n, and substituting $z \mapsto 1 - \rho + \rho z$ for independent reporting gives $[1 + (\rho R_t/k)(1 - z)]^{-nk}$, which proves Eq. (2); its derivatives give the stated moments.

For fixed, known (n, ρ, k) and $R > 0$, the count likelihood has score and Fisher information

$$\dot{\ell}(R) = \frac{c}{R} - \frac{(c + nk)\rho}{k + \rho R} = \mathcal{I}(R)(\hat{R} - R), \qquad \mathcal{I}(R) = \frac{m}{R + \rho R^2/k}. \tag{A.1}$$

By Eq. (A.1), $\mathrm{Var}(\hat{R} \mid R) = \mathcal{I}(R)^{-1}$. The support does not depend on R, so the regular one-parameter Cramér–Rao inequality applies to estimators unbiased for every R and gives the attained bound stated in Section 2.2 [23]. The bound therefore applies to estimators unbiased for every R with ρ and k known; biased estimators and unknown nuisance parameters require a separate analysis.

A.2 Proof of Proposition 1

Let $e_t = \hat{R}_t - R_t$. The error is conditionally centred, $\mathrm{E}(e_t \mid R_t) = 0$, and its second moments are as follows:

$$\mathrm{E}(e_t^2) = \frac{\mathrm{E}\, R_t + \rho\, \mathrm{E}\, R_t^2/k}{m} = \frac{A}{m}, \qquad \mathrm{E}[e_t(R_t - \overline{R})] = 0, \tag{A.2}$$

Squaring Eq. (4) proves Eq. (3). Any estimator based on the same count and unbiased conditional on every R_t has expected conditional variance at least $\mathrm{E}[\mathcal{I}(R_t)^{-1}] = A/m$, and its process-mean risk adds the same v_R. At fixed ρ, k and transmission law, $[V(m) - V(2m)]/V(m) = 1/[2(1 + m/m_\times)]$ for $v_R > 0$. The comparison changes exposure n; it does not hold unchanged when the reporting probability changes. The shrinkage weight in Eq. (5) is $\mathrm{Cov}(R_t, \hat{R}_t)/\mathrm{Var}(\hat{R}_t) = v_R/V(m)$.

A.3 Proof of Corollary 1

Conditional on the path $R_{1:T}$, the e_t are independent and centred, hence pairwise uncorrelated and orthogonal to every square-integrable function of the path. Thus the average of $e_1, \ldots, e_T$ has second moment $A/(Tm)$, and $\mathrm{Var}(\overline{\hat{R}}_T) = \mathrm{Var}(\overline{R}_T) + A/(Tm)$. Under the stated stationary autocovariance, summing all covariance pairs gives $T^2\,\mathrm{Var}(\overline{R}_T) = v_R S(\phi, T)$ with the first expression in Eq. (7), and the finite geometric sum yields the second [55]. Summing spatial covariance pairs in the same way gives Eq. (8), for any admissible correlation matrix.

A.4 Proof of Proposition 2 and Corollary 2

Write $z_t = p_t/\rho$ and $u_t = C_t/m - z_t R_t$. Conditional on (R_t, p_t), u_t has zero mean. Independence of p_t and R_t, with $\mathrm{E}\, z_t = 1$ and $\mathrm{E}\, z_t^2 = 1 + \tau^2$, gives

$$\mathrm{E}(u_t^2) = \frac{\overline{R} + \rho(1 + \tau^2)(\overline{R}^2 + v_R)/k}{m} = \frac{A_H}{m}. \tag{A.3}$$

The error $\hat{R}_t - R_t = u_t + (z_t - 1)R_t$ has uncorrelated terms, and the second has expected square $\tau^2(\overline{R}^2 + v_R) = v_H$, which proves Eq. (11). Because $\mathrm{E}(\hat{R}_t - R_t \mid R_t) = 0$, process-mean risk adds v_R. Under the additional cross-period assumptions of Corollary 2, the errors are centred and independent given the transmission path after integrating out reporting, which gives Eq. (12).

For the identification statement, set $U = (1 + \tau^2)(\overline{R}^2 + v_R)$. The two moments of the scaled count are

$$\mathrm{E}\, \hat{R}_t = \overline{R}, \qquad \mathrm{Var}(\hat{R}_t) = U - \overline{R}^2 + \frac{\overline{R} + \rho U/k}{m}. \tag{A.4}$$

Whenever distinct admissible values of τ^2 give $v_R = U/(1 + \tau^2) - \overline{R}^2 \geq 0$, the moments in Eq. (A.4) coincide, and varying an exogenous m does not separate them. This proves nonidentification by these moments only; it does not assert equality of full count distributions.

For $v_R > 0$, the crossover scale with volatile reporting, $m_{\times,H} = A_H/(v_R + v_H)$, satisfies

$$m_\times - m_{\times,H} = \frac{\tau^2(\overline{R}^2 + v_R)\overline{R}(1 + \rho\overline{R}/k)}{v_R(v_R + v_H)} \geq 0, \tag{A.5}$$

with equality exactly when $\tau = 0$.

A.5 Observed denominator

Let $M_t \sim \mathrm{Bin}(n, \rho)$ be independent of the offspring given n, and write $M_t = m(1+u_t)$, so that $\mathrm{E}\, u_t = 0$ and $\mathrm{Var}(u_t) = (1-\rho)/m$. Then $\tilde{R}_t = \hat{R}_t/(1+u_t)$. To first order in u_t and $\hat{R}_t - R_t$, the error is approximately:

$$\tilde{R}_t - R_t \approx (\hat{R}_t - R_t) - R_t u_t. \tag{A.6}$$

Given (n, R_t), u_t is independent of $\hat{R}_t - R_t$ and centred, so the cross term vanishes and $\mathrm{E}[(\tilde{R}_t - R_t)^2] \approx A/m + (1-\rho)\,\mathrm{E}(R_t^2)/m$, which is Eq. (9). The Monte Carlo check in Table 3 draws R_t from a gamma law with the stated moments and would discard draws with $M_t = 0$, none of which occurred at the scales shown. The approximation neglects the convexity of $1/(1+u_t)$, which inflates the error when m is small; at $m = 10$ the Monte Carlo error is 1.64 times the delta value.

A.6 Reporting shocks in the ratio statistic

Let the count of week t in a state be reported with probability ρz_t, where the shocks z_t are independent between weeks with $\log z_t \sim N(-s^2/2, s^2)$. To first order, the statistic in Eq. (15) satisfies $\log R_{A,t} \approx \text{const} + \kappa(g)\log g_t$ with $\kappa(g) = (g_\mu/7)/[1+\theta\log(g)/7]$, because $\alpha\theta = g_\mu$. The shocks enter the observed growth factor as $\log g_t = \log g_t^{\mathrm{tr}} + \eta_t$, with $\eta_t = \log z_{t+1} - \log z_t$ and g_t^{tr} the growth factor without shocks. Hence $\mathrm{Var}(\eta_t) = 2s^2$, and the relative variance that the shocks add to the ratio statistic is

$$\tilde{v}_H \approx 2\kappa^2 s^2, \tag{A.7}$$

which does not scale with count and therefore enters the intercept a. A reporting level that is constant over consecutive weeks of a state ($z_{t+1} = z_t$) gives $\eta_t = 0$ and cancels. Consecutive rows share one shock with opposite signs, so $\mathrm{Cov}(\eta_t, \eta_{t+1}) = -s^2$ and their correlation is $-1/2$. The affine fit uses squared deviations, and for jointly Gaussian terms a correlation of $-1/2$ becomes a correlation of $+1/4$ between their squares. Resampling single weeks does not preserve this serial dependence, so interval coverage was checked directly. Under S6 and S8, which contain such shocks, the intervals for a, b and b/a covered the Monte Carlo means of their estimators in at least 99.5% of 400 panels (Table 10). For the realised mean over T consecutive rows of one state the mean of the shock terms telescopes to $(\log z_{T+1} - \log z_1)/T$, so their contribution to the error of the realised mean falls as T^{-2} rather than T^{-1}. With κ near 0.5 for the primary interval and $s = 0.122$ as in generator S6, Eq. (A.7) gives a shock component of about half the observed intercept.

B Extensions

B.1 Forecasting and shrinkage

Under fixed, known reporting and the stationary covariance of Corollary 1, with the counting error at t centred given the transmission path, the best forecast of R_{t+1} of the form $\overline{R} + \lambda(\hat{R}_t - \overline{R})$ has

$$\lambda^* = \frac{\phi v_R}{v_R + A/m} = \phi\,\gamma(m), \qquad \mathrm{MSE}_{\mathrm{lin}} = v_R - \frac{\phi^2 v_R^2}{v_R + A/m}, \tag{B.1}$$

which follows from $\mathrm{Cov}(\hat{R}_t, R_{t+1}) = \phi v_R$. The forecast weight in Eq. (B.1) is the shrinkage weight discounted by persistence, and its limit risk is $(1-\phi^2)v_R$. For the process mean, shrinking $\hat{R}_t$ towards a fixed benchmark μ_0 with oracle weight $V(m)/[V(m) + (\mu_0 - \overline{R})^2]$ can give risk below v_R; this lies outside the conditionally unbiased class of Proposition 1, and estimating the weight requires its own risk analysis.

B.2 Allocation between periods, cases and coverage

Proposition 3 (Budget allocation between periods and cases)**.** *Under H1–H4, independent, identically distributed transmission periods, conditionally independent counts, fixed ρ and linear costs, let the budget be $\mathcal{B} = T(c_0 + c_1 m)$ with $c_0, c_1 > 0$, and treat $T \geq 1$ and $m > 0$ as continuous. If $v_R > 0$ and $\mathcal{B} > c_0$, process-mean risk $(c_0 + c_1 m)(v_R + A/m)/\mathcal{B}$ is strictly convex in m and is minimised at*

$$m^* = \min\{(c_0 m_\times/c_1)^{1/2},\ (\mathcal{B} - c_0)/c_1\}. \tag{B.2}$$

The optimum in Eq. (B.2) has the form of cluster-size allocation in survey sampling and multilevel design [12, 56]: cases per period should grow only with the square root of $m_\times$ and of the relative cost of a period. Integer designs compare the risk over feasible integer T with $n_T = \lfloor(\mathcal{B}/T - c_0)/(c_1\rho)\rfloor$. Under temporal dependence the variance term becomes $v_R S(\phi, T)/T^2$ and no closed-form optimum follows.

Corollary 3 (Coverage against completeness)**.** *Under H2–H5, let p_t be the participating fraction of ν independent reporting units, each participating with probability ρ, so that $\tau^2 = (1-\rho)/(\rho\nu)$. A proportional increase in completeness ρ reduces single-period process-mean risk more than the same proportional increase in exposure n exactly when $m > \rho^2(\nu-1)/k$.*

Write the single-period risk as $v_R + v_H + A_H/m$ with $v_H = \tau^2 Q$, $Q = \overline{R}^2 + v_R$ and $m = \rho n$, and differentiate with respect to $\log\rho$ and $\log n$. The difference of the two derivatives is

$$\frac{\partial\,\text{risk}}{\partial\log\rho} - \frac{\partial\,\text{risk}}{\partial\log n} = -\frac{Q}{kn}\left[\frac{kn+\rho}{\rho\nu} - 1\right]. \tag{B.3}$$

Because $Q/(kn) > 0$, the difference is negative, and a proportional increase in ρ lowers the risk more than the same increase in n, exactly when $kn + \rho > \rho\nu$, that is, when $m > \rho^2(\nu-1)/k$. The mechanism is assumed; observed facility participation alone does not imply it.

B.3 Number of seasons

Suppose $S \geq 2$ season effects are independent Gaussian with unknown mean and variance $v_S > 0$, and the count law given the effects does not depend on these parameters. Any estimator unbiased for v_S then has relative standard error at least $(2/(S-1))^{1/2}$. Observing the effects themselves gives at least as much information as the counts, and their sample variance is the minimum-variance unbiased estimator [11, 23]. With the four seasons analysed here the bound is $(2/3)^{1/2} = 0.82$, and additional admissions within the same seasons leave it unchanged. This bound concerns the precision with which the between-season variance component can be estimated; it is distinct from the power of a test of equal season intercepts, which depends on the size of the season differences relative to their standard errors and is reported in Section 5.3.

B.4 Thinning and the delta approximation

The thinning check of Eq. (18) rests on a delta approximation, so its exact expectation was computed on the observed counts. For each retention probability q the expected increment of the inverse-count slope was evaluated by summing over the binomial law of every thinned count (Table S5; Fig. S1A). The exact increment rate is 0.467 at $q = 0.8$, rises to 0.505 at $q = 0.3$ and falls to 0.480 at $q = 0.2$ and 0.384 at $q = 0.1$. The least-squares rate over all q is 0.408, or 0.917 b_{P}, against the Monte Carlo estimate 0.405 (0.366–0.441), or 0.909 (0.810–0.999) b_{P} (Table S6).

The pooled rate is dominated by the smallest retention probability. The value $q = 0.1$ carries 76% of the leverage in the least-squares rate (Fig. S1B), and without it the rate is 0.488, or 1.096 b_{P}. At $q = 0.1$ many thinned counts are small, and the delta approximation of the variance of a log thinned count fails there. At $C = 5$ and $q = 0.1$ the exact added variance of $\log(\text{Bin}(C, q) + 0.5)$ is 0.21 of the delta value (Table S7; Fig. S1C). The positive-part truncation of the fitted mean never binds. The shortfall of the pooled rate below b_{P} is therefore a property of the approximation at small thinned counts, not evidence about the data-generating process.

In the audit the ratio c/b_{P} lies in 0.95–1.00 under S0, 0.96–1.04 under S4, 0.87–0.98 under S5, 0.83–0.93 under S3 and 0.81–0.92 under S6. The observed value lies outside the ranges of the generators with Poisson offspring (S0, S1, S2P and S4) and inside those with negative binomial offspring, but its confidence interval overlaps all of them. Because thinning adds known binomial variance whatever the original counting process, the check cannot separate offspring overdispersion from under-reporting.

B.5 Cross-regime check on COVID-19 counts from mainland China

The diagnostics were applied to weekly confirmed COVID-19 cases from the Johns Hopkins CSSE repository [54] for the 31 provincial-level units of mainland China. The primary period runs from 25 January 2020 to 3 December 2022, before routine mass testing ended on 7 December 2022. Hong Kong SAR and Macao SAR enter only a sensitivity analysis, and Taiwan, China is not in the extract. Rows are unit-weeks with $C_t \geq 5$ inside national growth windows. Each window starts at a national peak of at least 100 weekly cases and extends back while the national growth factor is at least 1.05. The rule does not use the growth factor of the row itself, so the selection does not depend on the regressed outcome.

The generation interval is gamma with mean 3.95 days and standard deviation 1.51 days before 2022 [57]. From 2022 the Omicron BA.1 serial interval, with mean 3.5 days and standard deviation 2.4 days, serves as a proxy [58], and b_{P} uses the interval of each row. Negative daily increments, which occur in 30 unit-weeks, are set to zero.

The primary fit on 711 unit-weeks gave $a = 0.133$ (0.074–0.223), $b/a = 13.2$ (1.3–42.3) and $b/b_{\text{P}} = 0.54$ (0.05–2.78) (Table S16). The intercept is about nine times the influenza value. It stays between 0.119 and 0.146 for the full extract to 21 January 2023, window thresholds of 1.02 and 1.10, inclusion of the two SARs, removal of weeks with negative revisions, counts of at least 20 and the two sub-periods. Because a contains reporting variation (Proposition 2), the larger intercept is consistent both with greater variation in testing and case definitions and with greater transmission heterogeneity, and these

data do not separate the two. The slope-ratio interval contains 1 and values well below it, so the panel is uninformative about counting variation relative to the Poisson reference.

Two specification choices change the results. A single interval with mean 5.2 days throughout raises a to 0.298 and b/b_{P} to 4.62, and selecting rows on their own growth factor lowers a to 0.054 and raises b/b_{P} to 2.18. The comparison therefore supports one conclusion. Under every outcome-independent specification with a period-specific interval, the intercept in this regime is several times larger than in US influenza hospital admissions.

The comparison has further limits. Confirmed cases in this period may include imported cases, testing regimes and case definitions changed repeatedly, and the JHU CSSE series contains batch reporting, so weekly counts mix transmission with surveillance practice. The intercept also depends strongly on the generation interval, since a single 5.2-day interval more than doubles it, from 0.133 to 0.298. The panel is therefore a sensitivity check of the diagnostics in a different reporting regime and supports no conclusion about transmission.

C Supplementary tables and figures

Table S1 Known-truth audit over transmission share and scale dependence: compatibility of the observed statistics

γ	Share ϕ	Shock s	amp	a / v_R	a / (v_R + v_H)	b/b_P	b/a	Quantile of observed contrast (MC SE)	Observed statistics vs simulated range
0	0	0.172	0.977	v_R = 0	1.01	1.148 [0.944, 1.354]	36.5 [26.5, 49.6]	0.011 (0.003)	contrast outside
0	0.25	0.149	0.981	4.05	1.02	1.142 [0.947, 1.331]	36.1 [26.3, 48.2]	0.004 (0.002)	contrast outside
0	0.5	0.123	1.001	2.03	1.02	1.147 [0.949, 1.356]	35.5 [25.8, 46.8]	0.000 (0.000)	contrast outside
0	0.75	0.087	1.000	1.35	1.01	1.143 [0.946, 1.340]	35.7 [26.5, 45.9]	0.003 (0.002)	contrast outside
0	1	0.000	0.990	1.02	1.02	1.141 [0.970, 1.323]	36.0 [28.0, 45.4]	0.001 (0.001)	contrast outside
0.25	0	0.174	1.000	v_R = 0	1.02	1.148 [0.937, 1.360]	35.5 [25.2, 47.0]	0.007 (0.003)	contrast outside
0.25	0.25	0.156	1.074	3.85	0.94	1.212 [1.022, 1.430]	38.4 [27.9, 53.2]	0.026 (0.005)	all inside
0.25	0.5	0.135	1.200	1.75	0.86	1.297 [1.087, 1.521]	40.8 [30.1, 54.4]	0.070 (0.008)	all inside
0.25	0.75	0.100	1.323	1.04	0.77	1.415 [1.200, 1.644]	45.6 [34.5, 59.0]	0.165 (0.012)	all inside
0.25	1	0.000	1.513	0.69	0.69	1.558 [1.327, 1.779]	50.9 [38.0, 64.6]	0.462 (0.016)	b/b_P, b/a outside
0.5	0	0.174	0.998	v_R = 0	1.01	1.143 [0.944, 1.350]	35.6 [25.6, 48.2]	0.007 (0.003)	contrast outside
0.5	0.25	0.161	1.152	3.78	0.89	1.259 [1.048, 1.484]	39.9 [28.8, 53.9]	0.024 (0.005)	contrast outside
0.5	0.5	0.147	1.429	1.55	0.74	1.454 [1.220, 1.712]	45.9 [32.9, 63.2]	0.126 (0.010)	all inside
0.5	0.75	0.117	1.816	0.80	0.59	1.748 [1.467, 2.026]	57.2 [40.5, 76.4]	0.524 (0.016)	b/b_P, b/a outside
0.5	1	0.000	2.419	0.43	0.43	2.275 [1.978, 2.574]	81.3 [59.2, 106.5]	0.993 (0.003)	b/b_P, b/a, contrast outside
0.25, amp = 1	0	0.174	1.000	v_R = 0	1.01	1.147 [0.951, 1.367]	35.6 [25.8, 48.4]	0.007 (0.003)	contrast outside
0.25, amp = 1	0.25	0.150	1.000	3.86	0.94	1.211 [1.017, 1.417]	41.0 [29.7, 54.7]	0.018 (0.004)	contrast outside
0.25, amp = 1	0.5	0.123	1.000	1.74	0.86	1.290 [1.095, 1.501]	48.6 [35.7, 64.1]	0.046 (0.007)	a outside
0.25, amp = 1	0.75	0.087	1.000	1.03	0.77	1.375 [1.171, 1.562]	58.5 [43.4, 76.4]	0.155 (0.011)	b/a, a outside
0.25, amp = 1	1	0.000	1.000	0.67	0.67	1.484 [1.285, 1.684]	73.9 [57.3, 94.1]	0.323 (0.015)	b/b_P, b/a, a outside

The generator family is that of S8, with iid heterogeneity, negative-binomial offspring (k = 0.3), reporting probability 0.25, scale dependence γ and state-week reporting shocks of log-SD s. ϕ is the share of the total relative variance target carried by transmission; the shock carries 1 - ϕ (2 κ^2 s^2 = (1 - ϕ) amp a_{obs}). The cells with ϕ = 0.5 reproduce S6 (γ = 0) and S8 (γ = 0.25), and ϕ = 1 gives S3, S7a and S7b. Each row has 1000 panels (5 streams × 200). Rows without 'amp = 1' are calibrated, with amp chosen so that the MC mean of a equals the observed a = 0.0150. In those rows a is a target and only b/b_P, b/a and the contrast are tests. With amp = 1 (total variance target = observed a) a is tested as well. The last column records whether the observed b/b_P = 1.257, b/a = 37.4, contrast = -0.00136 (and a) fall inside the simulated 2.5-97.5% range. Ratios a / v_R and a / (v_R + v_H) are ratios of panel means.

Table S2 Share grid under the six declared tests, with the uncertainty in a propagated to the implied transmission relative SD

γ	ϕ	National	b/b_P	b/a	Thin.	Pooled/ within	Week centring	Contrast	Passed	Contrast quantile	Implied SD, %
0	0	0	✓	✓	✓	–	–	–	3/6	0.011	n.d.
0	0.25	0	✓	✓	✓	–	–	–	3/6	0.004	6.1 (5.2–6.8)
0	0.5	0	✓	✓	✓	–	–	–	3/6	0.000	8.6 (7.3–9.7)
0	0.75	0	✓	✓	✓	–	–	–	3/6	0.003	10.5 (8.9–11.9)
0	1	0	✓	✓	✓	–	–	–	3/6	0.001	12.1 (10.3–13.7)
0	0	0.5	✓	✓	–	–	–	–	2/6	0.014	n.d.
0	0.25	0.5	✓	✓	✓	–	–	–	3/6	0.009	6.0 (5.1–6.8)
0	0.5	0.5	✓	✓	–	–	–	–	2/6	0.003	8.4 (7.2–9.6)
0	0.75	0.5	✓	✓	–	–	✓	–	3/6	0.000	10.2 (8.7–11.6)
0	1	0.5	✓	✓	–	–	–	–	2/6	0.000	11.7 (10.0–13.3)
0.25	0	0	✓	✓	✓	–	–	–	3/6	0.007	n.d.
0.25	0.25	0	✓	✓	✓	–	–	✓	4/6	0.026	6.2 (5.3–7.0)
0.25	0.5	0	✓	✓	✓	–	–	✓	4/6	0.070	9.2 (7.9–10.4)
0.25	0.75	0	✓	✓	✓	–	–	✓	4/6	0.165	12.0 (10.3–13.5)
0.25	1	0	–	–	✓	–	–	✓	2/6	0.462	14.8 (12.8–16.5)
0.25	0	0.5	✓	✓	–	–	–	–	2/6	0.011	n.d.
0.25	0.25	0.5	✓	✓	✓	–	–	–	3/6	0.019	6.1 (5.2–6.9)
0.25	0.5	0.5	✓	✓	✓	–	–	✓	4/6	0.048	9.0 (7.7–10.2)
0.25	0.75	0.5	✓	✓	✓	–	✓	✓	5/6	0.111	11.5 (9.9–13.0)
0.25	1	0.5	✓	✓	✓	–	–	✓	4/6	0.160	13.9 (12.0–15.5)
0.5	0	0	✓	✓	✓	–	–	–	3/6	0.007	n.d.
0.5	0.25	0	✓	✓	✓	–	–	–	3/6	0.024	6.3 (5.3–7.1)
0.5	0.5	0	✓	✓	✓	–	–	✓	4/6	0.126	9.8 (8.4–11.1)
0.5	0.75	0	–	–	✓	–	–	✓	2/6	0.524	13.7 (11.8–15.2)
0.5	1	0	–	–	✓	–	–	–	1/6	0.993	18.7 (16.5–20.5)
0.5	0	0.5	✓	✓	–	–	–	–	2/6	0.007	n.d.
0.5	0.25	0.5	✓	✓	✓	–	–	✓	4/6	0.050	6.2 (5.3–7.0)
0.5	0.5	0.5	✓	✓	✓	–	–	✓	4/6	0.137	9.6 (8.1–10.8)
0.5	0.75	0.5	–	✓	✓	–	✓	✓	4/6	0.494	13.1 (11.3–14.4)
0.5	1	0.5	–	–	✓	–	–	–	1/6	0.980	16.9 (15.0–18.4)

γ, scale dependence of transmission heterogeneity; ϕ, share of the variance target carried by transmission, the rest by state-week reporting shocks; National, share of the transmission draw carried by a national calendar-week component. A tick marks an observed value inside the 2.5–97.5% Monte Carlo range of the cell, or for the contrast a quantile in $[0.025, 0.975]$ over 1,000 panels. Each cell has 200 panels, and the thinning test 100 panels with 20 draws. The numbers of cells passing each test are b/b_P 25, b/a 26, Thin. 24, Pooled/within 0, Week centring 3, Contrast 12, out of 30. The implied SD is the transmission relative SD implied by the observed a through the cell's ratio $a/\tilde{v}_R$, and its range recalibrates the cell to the limits of the two-way interval for a. n.d., no transmission heterogeneity.

Table S3 Pooled affine loss fits and generation-interval sensitivities (bootstrap 95% confidence interval in brackets)

Statistic	Parameter	n	Estimate [95% CI]	Finite/B
RA	a	1748	0.01495 [0.01006, 0.02063]	2000/2000
RA	b	1748	0.56 [0.362, 0.7694]	2000/2000
RA	b/a	1748	37.44 [18.41, 72.37]	2000/2000
G	a	1756	0.08401 [0.05023, 0.1243]	2000/2000
G	b	1756	1.971 [0.9495, 3.06]	2000/2000
G	b/a	1756	23.46 [8.046, 56.36]	2000/2000
RA32	a	1748	0.01191 [0.007987, 0.01641]	2000/2000
RA32	b	1748	0.4482 [0.2924, 0.6142]	2000/2000
RA32	b/a	1748	37.62 [18.72, 72.05]	2000/2000
RA32source	a	1748	0.01106 [0.007437, 0.01509]	2000/2000
RA32source	b	1748	0.4196 [0.2793, 0.5704]	2000/2000
RA32source	b/a	1748	37.93 [19.5, 71.31]	2000/2000

Pooled rows use the primary bootstrap (2,000 draws), and the same state-week weights are used for every statistic and generation interval. Intervals are signed percentile intervals, and ratios are defined only for positive denominators. Finite-draw counts are retained for every estimate. All 2,000 draws were finite for a, b and their ratio under RA, G, RA32 and RA32source.

Table S4 Branching-Poisson and independent-Poisson slope references, analytic, simulated and observed

Reference	Analytic reference / b_P	Simulated b/b_P (no heterogeneity)	MC SE	Simulated / analytic	Observed b / reference
Branching-Poisson, ρ = 0.1	0.944	0.877 [0.802, 0.955]	0.0029	0.929	1.33 [0.86, 1.84]
Branching-Poisson, ρ = 0.25	0.861	0.802 [0.734, 0.878]	0.0026	0.932	1.46 [0.94, 2.02]
Branching-Poisson, ρ = 0.5	0.722	0.669 [0.618, 0.714]	0.0019	0.927	1.74 [1.12, 2.41]
Branching-Poisson, ρ = 1	0.443	0.402 [0.373, 0.434]	0.0012	0.907	2.84 [1.83, 3.94]
Independent-Poisson b_P	1	0.930 [0.853, 1.005]	0.0029	0.930	1.26 [0.81, 1.73]

The branching-Poisson reference is $b_{P,\mathrm{br}}(\rho)$ = mean over the observed retained rows of κ^2 [(1 - ρ) + (C_t + 0.5)/(C_{t+1} + 0.5)]; independent-Poisson reference b_P = 0.4455. Simulated rows use a pure branching chain with Poisson offspring and binomial reporting ρ (independent-Poisson counts, S0, in the last row), no transmission heterogeneity, count-aligned as in the audit. Each has 200 panels, with the Monte Carlo 2.5-97.5% range in brackets, and simulated / analytic is not normalised by the independent-Poisson row. The observed row uses the primary bootstrap (2,000 draws), with $b_{P,\mathrm{br}}$ recomputed with the weights of each draw. The ratio of the observed slope to the reference depends on the reporting probability.

Table S5 Exact expected thinning increment against the delta prediction b_P

q	Exact rate	Exact rate / b_P	MC rate [95% CI]	Leverage in c	Positive-part mass (max)
0.8	0.4672	1.049	0.4588 [0.3731, 0.5557]	0.001	0.0e+00
0.6	0.4845	1.087	0.4745 [0.4064, 0.5532]	0.004	0.0e+00
0.5	0.4957	1.113	0.4969 [0.4243, 0.5606]	0.009	0.0e+00
0.4	0.5050	1.133	0.4997 [0.4397, 0.5611]	0.021	0.0e+00
0.3	0.5051	1.134	0.5067 [0.4482, 0.5539]	0.051	0.0e+00
0.2	0.4798	1.077	0.4749 [0.4292, 0.5229]	0.151	0.0e+00
0.1	0.3836	0.861	0.3804 [0.3337, 0.4204]	0.763	0.0e+00
c, all q	0.4084	0.917	0.4051 [0.3656, 0.4413]	1.000	–
c, without q = 0.1	0.4882	1.096	–	–	–

Rate = (b_q - b_1)/(1/q - 1) from the affine FGLS fit of the exact expected thinned loss (binomial supports summed, mass below 1e-13 dropped); b_P = 0.4455 is the delta-method prediction. Leverage is the share of the least-squares rate through the origin carried by each q. As a check, independent-thinning MC gives 0.4903 (SE 0.0048) against an expected-loss value of 0.4957 at q = 0.5, and 0.3832 (SE 0.0024) against 0.3836 at q = 0.1.

Table S6 Shared-count binomial thinning of the R_A sample (n=1748): summary by subset

Subset (n)	Fit over	b_P [95% CI]	c [95% CI]	c/b_P [95% CI]	Δa rate $\times 10^3$ [95% CI]	c/b_P by q, range
full (1748)	all q	0.446 [0.428, 0.466]	0.405 [0.366, 0.441]	0.909 [0.810, 0.999]	-0.13 [-0.62, 0.47]	0.85–1.14
	$q \geq 0.2$		0.485 [0.439, 0.528]	1.088 [0.994, 1.186]	-0.81 [-1.70, 0.05]	1.03–1.14
highCt50 (828)	all q	0.421 [0.403, 0.439]	0.473 [0.409, 0.550]	1.126 [0.970, 1.316]	-0.19 [-0.69, 0.14]	1.05–1.14
	$q \geq 0.2$		0.459 [0.381, 0.560]	1.092 [0.903, 1.327]	-0.24 [-0.96, 0.21]	1.05–1.10
both50 (821)	all q	0.418 [0.401, 0.436]	0.493 [0.420, 0.574]	1.178 [1.004, 1.379]	-0.31 [-0.90, 0.10]	1.16–1.21
	$q \geq 0.2$		0.490 [0.397, 0.607]	1.171 [0.953, 1.462]	-0.43 [-1.33, 0.12]	1.16–1.21

Thinning probabilities $q \in \{0.8, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1\}$. c is the increase of the slope per unit of $1/q - 1$ and the Δa rate the corresponding change of the intercept; b_P is the independent-Poisson reference (plug-in delta value). highCt50 requires a current count of at least 50 and both50 both adjacent counts of at least 50; eligibility is fixed before thinning. Pairs bootstrap with B=1000 draws, 400 thinning replicates for the point estimate and 50 per bootstrap draw; the Monte Carlo precision criterion (MC SE at most 0.1 times the point estimate on the monitored summaries) is met; point fits 8400/8400, bootstrap fits 1049998/1050000. The last column gives the range of the per-q point estimates of c/b_P over the q values of the fit. Per-q estimates, simultaneous bands over q and leave-one-q-out fits are in the released file analysis/authors/thinning.json.

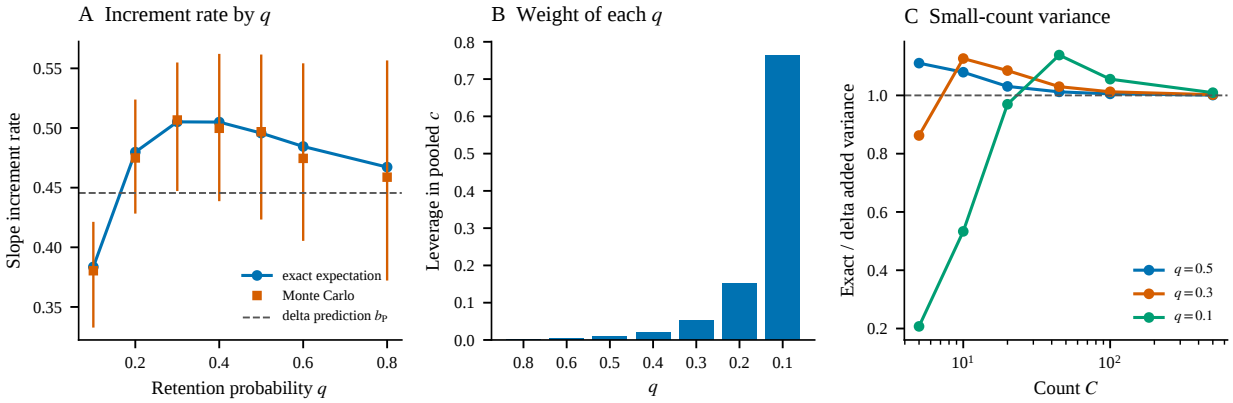


Figure S1 Exact thinning expectation on the observed counts. Panel A shows the expected increment rate of the inverse-count slope by retention probability q, exact expectation against Monte Carlo estimates with 95% ranges; the dashed line is the delta prediction $b_P = 0.4455$. Panel B shows the share of each q in the leverage of the pooled least-squares rate c. Panel C shows the exact added variance of the log thinned count relative to its delta approximation, by original count C, for three retention probabilities.

Table S7 Small-count variance of the log thinned count relative to the delta approximation

C	q = 0.8	q = 0.6	q = 0.5	q = 0.4	q = 0.3	q = 0.2	q = 0.1
5	1.05	1.11	1.11	1.04	0.86	0.57	0.21
10	1.02	1.05	1.08	1.12	1.13	0.99	0.53
20	1.01	1.02	1.03	1.05	1.09	1.14	0.97
45	1.00	1.01	1.01	1.02	1.03	1.06	1.14
100	1.00	1.00	1.01	1.01	1.01	1.02	1.06
500	1.00	1.00	1.00	1.00	1.00	1.00	1.01

Exact added variance Var log(Bin(C, q) + 0.5) - Var log(C + 0.5) divided by the delta value (1 - q)/(qC). Values below 1 mean that thinning adds less variance than the delta method assumes.

Table S8 Centering stages with a common loss normalizer

Statistic / center	Pooled mean	a / change [95% CI]	b [95% CI]	Finite/B
RA / pooled	1.206	0.01495 [0.01006, 0.02063]	0.56 [0.362, 0.7694]	2000/2000
RA / season	1.206	0.01471 [0.009507, 0.01966]	0.5391 [0.3602, 0.7205]	1000/1000
RA / season-linear	1.206	0.0154 [0.008972, 0.01901]	0.3936 [0.2717, 0.5763]	1000/1000
RA / week	1.206	0.009766 [0.007103, 0.01357]	0.4581 [0.3124, 0.5916]	1000/1000
RA / week, leave-state-out	1.206	0.01016 [0.007654, 0.01481]	0.4841 [0.3492, 0.6636]	1000/1000
RA / week change	1.206	-0.005189 [-0.01015, -0.0006938]	–	1000/1000
RA / week reduction fraction	1.206	34.7 [5.967, 51.9]	–	1000/1000
G / pooled	1.724	0.08401 [0.05023, 0.1243]	1.971 [0.9495, 3.06]	2000/2000
G / season	1.724	0.08163 [0.0504, 0.1156]	1.842 [0.9591, 2.602]	1000/1000
G / season-linear	1.724	0.07864 [0.04808, 0.09897]	1.249 [0.6423, 1.988]	1000/1000
G / week	1.724	0.05607 [0.03962, 0.07812]	1.401 [0.7134, 1.912]	1000/1000
G / week, leave-state-out	1.724	0.05843 [0.04308, 0.08505]	1.477 [0.8089, 2.138]	1000/1000
G / week change	1.724	-0.02794 [-0.05687, -0.003708]	–	1000/1000
G / week reduction fraction	1.724	33.25 [6.805, 51.63]	–	1000/1000

All five centers use the same pooled mean as the relative-loss denominator within each statistic; the denominator is refitted with weights within each bootstrap draw. week, leave-state-out removes the focal state from its weekly center. Reduction is $100*(a_{\text{pooled}}-a_{\text{week}})/a_{\text{pooled}}$, in percent; changes are descriptive and not additive variance components. Pooled-centre rows use the primary bootstrap (2,000 draws, as in Table S3); the other centres and the week change use this table's own 1000-draw bootstrap. Every draw was finite, 2,000 of 2,000 for the pooled centre and 1,000 of 1,000 for each other centre, for both RA and G.

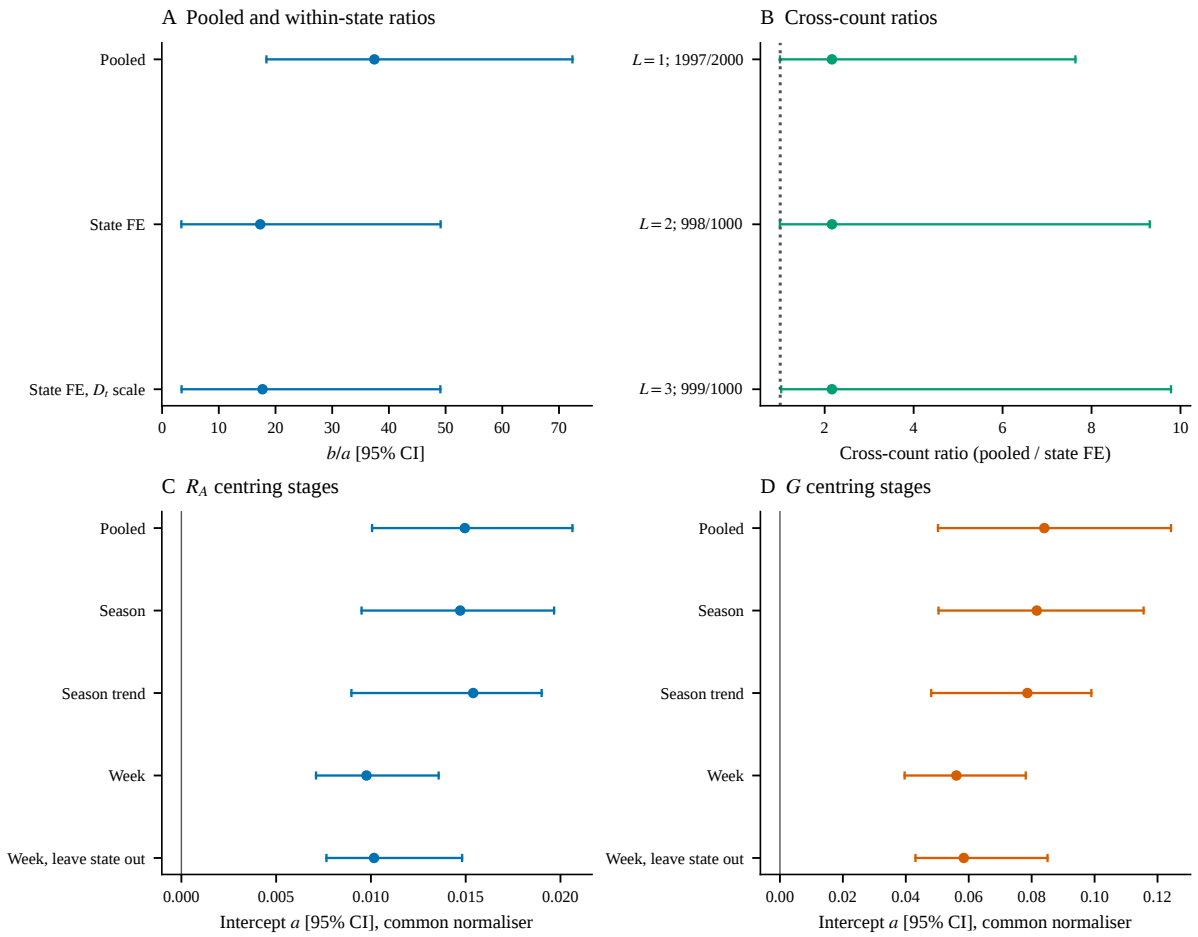


Figure S2 Heterogeneity diagnostics for the R_A and G samples. Panel A shows the crossover count b/a for R_A from the pooled fit, a state fixed-effects fit and a state fixed-effects fit with D_t as count scale. Panel B shows the cross-count ratio, pooled b/a over state fixed-effects b divided by the mean state intercept, under week resampling ($L = 1$) and two- and three-week blocks ($L = 2, 3$), with the number of finite bootstrap draws; the dotted line marks 1. Panels C and D show the intercept a under five centring choices for R_A and G, each with the pooled mean of its statistic as common normaliser. Bars are bootstrap 95% confidence intervals.

Table S9 Season-specific fits and reporting participation sensitivity

Statistic / period	n	a / change [95% CI]	b / original a	b/a / adjusted a	Finite/B
RA / 2022-23	365	0.01369 [0.0071, 0.02102]	0.6967	50.91	1000/1000
RA / 2023-24	499	0.006174 [0.002667, 0.01542]	0.7018	113.7	1000/1000
RA / 2024-25	356	0.01268 [0.005687, 0.02276]	0.5826	45.93	1000/1000
RA / 2025-26	528	0.02253 [0.008432, 0.03509]	0.3533	15.68	1000/1000
RA / reporting adjustment	1748	0.0002939 [-0.0003183, 0.0008476]	0.01495	0.01466	1000/1000
G / 2022-23	367	0.07179 [0.03786, 0.1162]	2.702	37.64	1000/1000
G / 2023-24	503	0.03046 [0.0118, 0.07462]	2.382	78.2	1000/1000
G / 2024-25	357	0.06469 [0.03191, 0.1278]	2.082	32.19	1000/1000
G / 2025-26	529	0.1383 [0.03715, 0.2252]	0.7411	5.359	1000/1000
G / reporting adjustment	1756	0.000373 [-0.003559, 0.003454]	0.08401	0.08364	1000/1000

Season rows report a with its interval and b, b/a as point estimates only. Reporting-adjustment rows use the a column for original-minus-adjusted a [95% CI], b for original a, and b/a for adjusted a; the original normalizer is fixed. Season b and b/a come from separate per-season fits without a stored bootstrap; season b/a with intervals from a common-slope model are given in the season table. Summer rows are reported in a separate table. All 1,000 draws were finite in every season for RA and G. Season intervals here use 1,000 draws and those of Table 12 2,000, so their limits differ slightly.

Table S11 Cross-count ratios, slope ratios and intercept ratios for the R_A sample ($n = 1,748$), with week-block sensitivity

Block Length	Parameter	Estimate [95% CI]	Valid Draws	Diagnostic Note
$L = 1$	Pooled a	0.01495 [0.01006, 0.02063]	2000/2000	Converged
$L = 1$	Pooled b	0.560 [0.3620, 0.7694]	2000/2000	Converged
$L = 1$	Pooled b/a	37.44 [18.41, 72.37]	2000/2000	Primary bootstrap
$L = 1$	State FE b	0.3794 [0.1069, 0.6502]	2000/2000	Within-state slope
$L = 1$	State FE mean a	0.02192 [0.01266, 0.03544]	2000/2000	Mean state intercept
$L = 1$	State FE b/mean a	17.30 [3.402, 49.12]	2000/2000	Nonpositive frac. 0.0015
$L = 1$	Slope ratio (pooled b / state FE b)	1.476 [0.9883, 3.662]	1997/2000	Spans 1.0 (unstable denom.)
$L = 1$	Intercept ratio (state FE mean a / pooled a)	1.466 [0.9892, 2.291]	2000/2000	Spans 1.0
$L = 1$	Cross-count ratio (pooled b/a / state FE b/mean a)	2.164 [0.9946, 7.637]	1997/2000	Ratio of cross counts
$L = 2$	State FE b	0.3794 [0.1045, 0.6646]	1000/1000	2-week block bootstrap
$L = 2$	State FE mean a	0.02192 [0.01245, 0.03616]	1000/1000	
$L = 2$	Slope ratio (pooled b / state FE b)	1.476 [0.9922, 4.290]	998/1000	
$L = 2$	Cross-count ratio	2.164 [0.9939, 9.309]	998/1000	
$L = 3$	State FE b	0.3794 [0.09549, 0.6499]	999/1000	3-week block bootstrap
$L = 3$	State FE mean a	0.02192 [0.01315, 0.03666]	999/1000	
$L = 3$	Slope ratio (pooled b / state FE b)	1.476 [1.028, 4.598]	999/1000	
$L = 3$	Cross-count ratio	2.164 [1.023, 9.787]	999/1000	
Diagnostic	Week-only slope b	0.5258	Point fit	Objective fallback
Diagnostic	State and week two-way FE	Unavailable	–	Iteration limit

The sample is the R_A specification, $n = 1,748$ state-weeks in 51 jurisdictions. FE targets sampled state occurrences; pooled and FE quantities are computed on the same draw. Cross-count ratio = slope ratio × intercept ratio. L denotes within-season week-block length. Week-only and state-and-week fits are point diagnostics. Full listings across all alternative weightings and estimators are archived in the supplementary materials.

Table S10 Within-state, within-week and two-way slopes against the independent-Poisson reference (bootstrap 95% confidence interval in brackets)

Estimator	Slope [95% CI]	Slope / b_P	Slope / pooled b	Fit status
Pooled affine slope b	0.560 [0.362, 0.769]	1.26 [0.81, 1.73]	–	–
b_P, independent-Poisson reference	0.446 [0.427, 0.465]	–	–	–
State FE, iterated FGLS (b_w)	0.379 [0.107, 0.650]	0.85 [0.24, 1.47]	0.68 [0.26, 1.01]	–
State FE, one-step WLS	0.456 [0.228, 0.682]	1.02 [0.51, 1.53]	0.81 [0.57, 0.96]	single step
Week FE, one-step WLS	0.634 [0.420, 0.846]	1.42 [0.96, 1.89]	1.13 [0.98, 1.34]	single step
State + week FE, one-step WLS	0.480 [0.208, 0.744]	1.08 [0.47, 1.66]	0.86 [0.47, 1.14]	single step
State + week FE, damped iterated FGLS	0.414 [0.040, 0.659]	0.93 [0.09, 1.48]	0.74 [0.09, 1.10]	converged, 124 it.

The primary bootstrap uses the state-week weights of the primary stream with 2,000 draws, so the pooled row equals Table S3; draws whose iterated fit does not converge are dropped (1962 valid for the damped two-way slope). The undamped two-way iteration did not converge after 51 iterations (last values 0.4150, 0.4122); damping 0.5 removes the two-cycle. One-step WLS uses the pooled affine working precision; the iterated estimators re-weight with the fixed-effect fitted means.

Table S12 Unthinned slope relative to the independent-Poisson reference over reporting probability and offspring dispersion

Reporting probability ρ	Branching r(ρ)	k = 0.1	k = 0.3	k = 1	k = 3	Poisson offspring
0.1	0.944	1.28 [1.11, 1.47]	0.98 [0.86, 1.13]	0.86 [0.69, 1.01]	0.82 [0.69, 0.99]	0.81 [0.68, 0.94]
0.25	0.861	1.80 [1.59, 2.03]	1.14 [0.95, 1.31]	0.86 [0.71, 1.02]	0.79 [0.67, 0.93]	0.76 [0.61, 0.89]
0.5	0.722	2.40 [2.07, 2.74]	1.41 [1.24, 1.59]	0.85 [0.69, 0.98]	0.71 [0.59, 0.81]	0.62 [0.50, 0.75]
1	0.443	2.89 [2.41, 3.34]	1.80 [1.53, 2.04]	0.88 [0.74, 1.04]	0.55 [0.41, 0.68]	0.37 [0.28, 0.48]

Unthinned b/b_P, with b_P the independent-Poisson reference, under S1 iid heterogeneity (true relative variance = observed a) with negative-binomial offspring of dispersion k and binomial reporting ρ. Each cell has 100 panels, with the Monte Carlo 2.5-97.5% range in brackets. r(ρ) = $b_{P,\mathrm{br}}(\rho)/b_P$ is the analytic branching-Poisson reference on the observed rows (no heterogeneity). Observed b/b_P = 1.26 [0.81, 1.73] (primary bootstrap, 2,000 draws).

Table S13 Season-stratified loss model with a common slope and tests of equal season intercepts (bootstrap 95% confidence interval in brackets)

Quantity	Pooled loss, season intercepts	Season-centred loss, season intercepts
Common slope b	0.607 [0.415, 0.794]	0.586 [0.400, 0.760]
a, 2022-23	0.0159 [0.0087, 0.0269]	0.0162 [0.0085, 0.0263]
a, 2023-24	0.0071 [0.0036, 0.0143]	0.0064 [0.0027, 0.0128]
a, 2024-25	0.0154 [0.0071, 0.0276]	0.0141 [0.0069, 0.0225]

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Quantity	Pooled loss, season intercepts	Season-centred loss, season intercepts
a, 2025-26	0.0180 [0.0075, 0.0278]	0.0185 [0.0057, 0.0277]
b/a, 2022-23	38.3 [17.7, 77.4]	36.2 [17.9, 75.4]
b/a, 2023-24	85.7 [31.8, 193.7]	91.5 [34.5, 228.0]
b/a, 2024-25	39.5 [17.5, 97.6]	41.7 [20.3, 94.6]
b/a, 2025-26	33.7 [17.1, 86.8]	31.7 [17.0, 108.0]
Wald test of equal log a_s (df 3)	W = 5.52, p = 0.138	–
Cochran Q, log a_s (df 3)	Q = 4.57, p = 0.206, I2 = 34%	–
Wald test of equal a_s (df 3)	W = 7.25, p = 0.064	–
Wald test, separate season fits, log b_s/a_s (df 3)	W = 4.41, p = 0.220	–
Minimum detectable highest/lowest a (80% power)	3.71 (observed 2.54)	–

Affine loss with one intercept per season and a common slope (group FGLS); state-week bootstrap, 2,000 draws (1,999 valid). The season-centred column centres $\times$ on season means before forming the loss. Tests use the bootstrap covariance of the season parameters (1,995 of 2,000 draws with all parameters finite); with a common slope, equal log a_s is equivalent to equal season b/a. Cochran Q uses inverse-variance weights. The separate-fit test refits the loss in each season on its own bootstrap stream. For the minimum detectable difference, noncentrality 10.90 gives 80% power for a level-0.05 chi-square(3) test; scaling the observed log a_s deviations by f = 1.41 reaches it, which corresponds to the highest/lowest a shown. The highest to lowest season b/a ratio (2.54 [1.52, 6.01]) is descriptive and is not a homogeneity test.

Table S14 Off-season summer descriptors

Period	n	Median count scale	Median participation (%)	IQR(log G)
Summer 2023	772	15.55	94.24	0.4873
Summer 2024	400	17.12	45.59	0.6683
Summer 2025	839	16.25	92.31	0.4882

Descriptive summaries of May-October state-weeks with $D_t \geq 5$, giving rows, median count scale D_t, median reporting participation (%) and IQR of log G. No model is fitted, so no interval is given. Reporting was voluntary in the 2024 summer only; that contrast is descriptive, not a causal or independently replicated intervention.

Table S15 Generation-interval sensitivity of the pooled loss model (bootstrap 95% confidence interval in brackets)

Generation interval	n	a	b	b/a	b_P	b/b_P	Paired change in b/a	Paired change in b/b_P
Gamma 3.6 / 1.6 (primary)	1748	0.0150 [0.0101, 0.0206]	0.560 [0.362, 0.769]	37.4 [18.4, 72.4]	0.446 [0.427, 0.465]	1.26 [0.81, 1.73]	reference	reference
Gamma 3.2 / 1.42	1748	0.0119 [0.0080, 0.0164]	0.448 [0.292, 0.614]	37.6 [18.7, 72.1]	0.354 [0.340, 0.368]	1.27 [0.83, 1.74]	0.17 [-0.65, 0.89]	0.010 [-0.005, 0.028]
Gamma 3.2 / 2.1	1748	0.0111 [0.0074, 0.0151]	0.420 [0.279, 0.570]	37.9 [19.5, 71.3]	0.340 [0.322, 0.360]	1.23 [0.83, 1.68]	0.49 [-2.55, 3.04]	-0.023 [-0.082, 0.036]

State-week bootstrap with 2000 draws; the same weights are reused for every generation interval (paired = True), so the last two columns are paired differences from the primary specification. Gamma 3.2 / 1.42 keeps the primary coefficient of variation; 3.2 / 2.1 is the intrinsic generation-time mean and SD estimated from US household data (Chan et al., 2025).

Table S16 China COVID-19 cross-regime check with outcome-independent windows (bootstrap 95% confidence interval in brackets)

Specification	Rows	Units	a	b/a	b/b_P	Share of draws with a ≤ 0
United States influenza (primary, for comparison)	1748	–	0.0150 [0.0101, 0.0206]	37.4 [18.4, 72.4]	1.26 [0.81, 1.73]	–
Primary, 31 mainland units, national windows (1.05), to 2022-12-03	711	31	0.133 [0.074, 0.223]	13.2 [1.3, 42.3]	0.54 [0.05, 2.78]	0.000
Full extract to 2023-01-21	793	31	0.146 [0.082, 0.240]	12.4 [1.4, 39.1]	0.59 [0.07, 3.00]	0.000
Window rule 1.10	592	31	0.135 [0.074, 0.229]	12.7 [0.6, 48.0]	0.49 [0.02, 2.91]	0.000
Window rule 1.02	758	31	0.144 [0.087, 0.249]	11.7 [1.1, 35.3]	0.54 [0.04, 2.40]	0.000
With Hong Kong SAR and Macao SAR	777	33	0.130 [0.078, 0.210]	13.7 [1.7, 43.5]	0.58 [0.06, 2.57]	0.000
Generation interval 5.2 / 1.7 d throughout	711	31	0.298 [0.100, 0.700]	27.5 [-0.3, 141.4]	4.62 [-0.08, 9.41]	0.000
Weeks with negative revisions dropped	707	31	0.133 [0.084, 0.223]	13.4 [1.4, 37.1]	0.55 [0.06, 2.59]	0.000
Counts ≥ 20	404	31	0.142 [0.056, 0.240]	9.1 [-7.3, 80.4]	0.29 [-1.09, 4.52]	0.000
Before 2022 only	289	30	0.132 [0.052, 0.317]	10.9 [-2.6, 66.3]	1.29 [-0.60, 3.87]	0.002
2022 only	422	31	0.119 [0.058, 0.221]	18.3 [1.3, 72.3]	0.47 [0.03, 3.38]	0.000
Own-row growth rule g ≥ 1.05 (selects on the outcome)	648	31	0.054 [0.028, 0.107]	14.8 [-1.6, 48.1]	2.18 [-0.40, 4.61]	0.000

Data source, units, window rule and generation intervals are as described in Appendix B.5; entries without a unit are excluded. Unit-week pigeonhole bootstrap, 1,000 draws (primary) and 500 (other rows).

Table S17 Held-out comparison of the affine and power loss models with paired uncertainty

Held-out unit	Folds	Mean power - affine score	Paired fold SE	z	Folds favouring affine	Bootstrap 95% CI	Bootstrap share favouring affine
State	51	-0.0020	0.0068	-0.30	26/51	[-0.0290, 0.0298]	0.43
Season	4	0.0046	0.0087	0.53	3/4	[-0.0295, 0.0655]	0.54

The score is the weighted mean of the γ quasi-deviance term $L_{st}/\hat{\mu}_{st} + \log \hat{\mu}_{st}$ over held-out rows, where $\hat{\mu}_{st}$ is the fitted mean of L_{st}; positive differences favour the affine model. Paired fold SE from the between-fold variation of the paired differences; state-week bootstrap with 1000 draws refits both models in every fold.

Table S18 Held-out power-minus-affine score contrast on synthetic panels from each generator

	Leave one state out (51 folds)			Leave one season out (4 folds)		
Generator	Mean contrast ($\times 10^3$)	Affine better	Observed quantile	Mean contrast ($\times 10^3$)	Affine better	Observed quantile
Observed	-2.02			4.63		
S0	-1.01 (0.19)	0.34	0.235 (0.030)	-0.89 (0.23)	0.38	0.970 (0.012)
S1	12.84 (0.37)	0.99	0.000 (0.000)	12.97 (0.37)	0.99	0.060 (0.017)
S2	-0.21 (0.13)	0.51	0.125 (0.023)	-0.09 (0.15)	0.50	0.985 (0.009)
S2P	-1.00 (0.18)	0.35	0.215 (0.029)	-0.81 (0.31)	0.42	0.980 (0.010)
S3	17.38 (0.44)	1.00	0.000 (0.000)	17.49 (0.44)	1.00	0.000 (0.000)
S4	11.21 (0.36)	0.99	0.000 (0.000)	15.78 (0.38)	1.00	0.010 (0.007)
S5	24.03 (0.48)	1.00	0.000 (0.000)	28.32 (0.58)	1.00	0.000 (0.000)
S6	18.74 (0.59)	0.99	0.000 (0.000)	18.60 (0.58)	0.99	0.045 (0.015)
S7a	-1.32 (0.44)	0.41	0.475 (0.035)	-0.85 (0.44)	0.44	0.805 (0.028)
S7b	-15.42 (0.38)	0.00	0.995 (0.005)	-14.31 (0.40)	0.01	1.000 (0.000)
S8	11.07 (0.60)	0.91	0.050 (0.015)	11.15 (0.62)	0.91	0.220 (0.029)
S9	11.83 (0.59)	0.92	0.055 (0.016)	12.60 (0.62)	0.93	0.145 (0.025)

200 synthetic panels per generator, each analysed with the fold design and score of the data: mean of $L/\mu + \log \mu$ on the held-out rows, weighted by rows, power law minus affine (positive values favour the affine form). Mean contrast with its Monte Carlo standard error; affine better, share of panels with a positive contrast; observed quantile, share of panels with a contrast at or below the observed one, with binomial Monte Carlo standard error.

Table S19 Foldwise model validation

Statistic / split	Fold	Valid/total or n	Affine	Power	Difference	Folds favouring affine
RA / state	Summary	51/51	–	–	-0.002018	26
RA / season	Summary	4/4	–	–	0.004626	3
RA	2022-23	365	-2.378	-2.369	0.008954	–
RA	2023-24	499	-2.613	-2.612	0.0008806	–
RA	2024-25	356	-2.46	-2.481	-0.02077	–
RA	2025-26	528	-2.346	-2.324	0.0223	–
G / state	Summary	51/51	–	–	-0.007369	22
G / season	Summary	4/4	–	–	-0.004414	2
G	2022-23	367	-0.5979	-0.5942	0.003756	–
G	2023-24	503	-1.201	-1.192	0.008487	–
G	2024-25	357	-0.7333	-0.7584	-0.02505	–
G	2025-26	529	-0.7557	-0.7641	-0.008421	–
RA32 / state	Summary	51/51	–	–	-0.002242	25
RA32 / season	Summary	4/4	–	–	0.004343	3
RA32	2022-23	365	-2.612	-2.604	0.00805	–
RA32	2023-24	499	-2.83	-2.83	0.0001619	–
RA32	2024-25	356	-2.691	-2.712	-0.02118	–
RA32	2025-26	528	-2.572	-2.549	0.02294	–
RA32source / state	Summary	51/51	–	–	-0.002824	25
RA32source / season	Summary	4/4	–	–	0.003577	2
RA32source	2022-23	365	-2.701	-2.695	0.005641	–
RA32source	2023-24	499	-2.886	-2.888	-0.001735	–
RA32source	2024-25	356	-2.77	-2.792	-0.02217	–
RA32source	2025-26	528	-2.644	-2.619	0.02453	–

Summary rows show valid/total folds and the number favouring affine in the last column; season rows show test n. All fold fits converged. The score is the mean of $L_{st}/\hat{\mu}_{st} + \log \hat{\mu}_{st}$, and lower is better. The difference is power minus affine, so positive values favour the affine model. The summary difference is the mean across folds weighted by held-out test n; the unweighted fold mean and the count of folds favouring affine are shown separately. Fold fits refit the training center. Training sets overlap; no iid-fold SE or four-season significance test is reported.

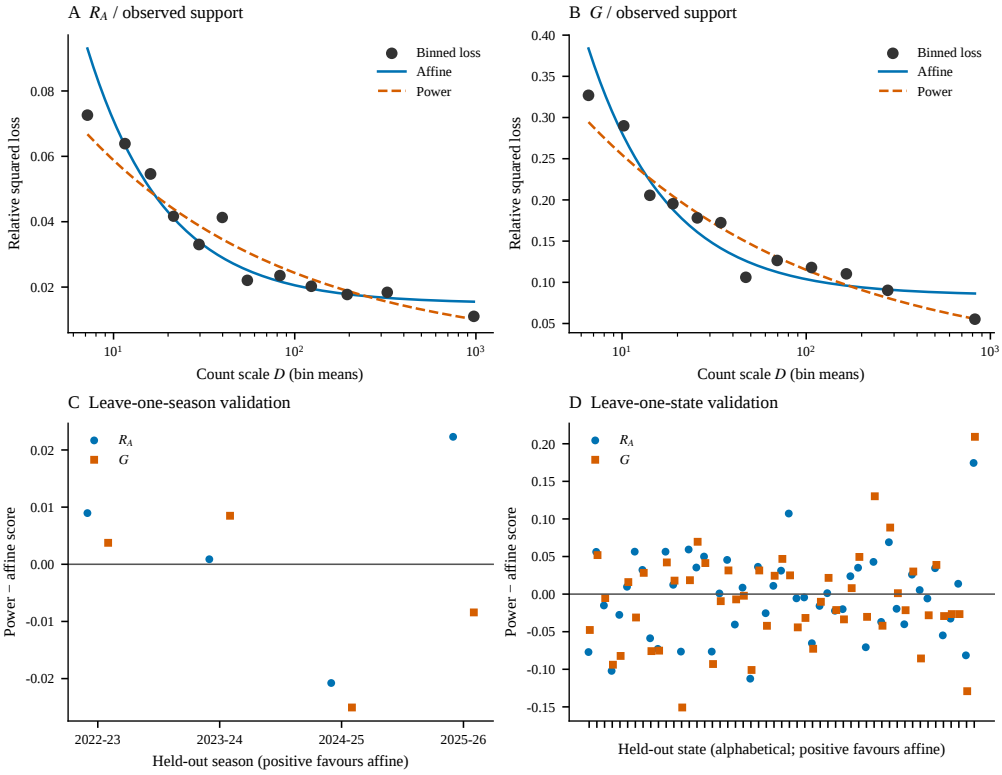


Figure S3 Affine and power-law forms of the dispersion curve. (A, B) Relative squared deviation averaged within count bins for R_A and G, with the fitted affine and power-law curves; bins are for display only. (C) Held-out power-minus-affine score difference for each left-out season. (D) The same for each left-out state. Positive differences favour the affine model.

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